EC442: Lecture Notes

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Notes on LSE EC442: Macroeconomics.

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1. Lecture 1 Sep. 29

1.1. Why heterogeneity

Complete competitive markets under the welfare-theorem assumptions:

- Equilibrium allocations are Pareto efficient. Consider an interior planner allocation with fixed welfare weights $\alpha_i > 0$.
- Common discounting; preferences $u(c) - v(n)$. Consumption FOCs:

$$\alpha_i u'\left(c_t^i\right) = \alpha_j u'\left(c_t^j\right) \Rightarrow \frac{u'\left(c_t^i\right)}{u'\left(c_t^j\right)} = \frac{\alpha_j}{\alpha_i}. \tag{1.1}$$

  Marginal-utility ratio constant across time/states.
- Common CRRA: $u'(c) = c^{-\sigma}$.

$$\frac{c_t^i}{c_t^j} = \left(\frac{\alpha_i}{\alpha_j}\right)^{\frac{1}{\sigma}}. \tag{1.2}$$

  Fixed consumption shares → **perfect insurance** against idiosyncratic income shocks. Aggregate risk remains; unequal Pareto weights → unequal consumption.
- With strictly convex labor disutility and an interior choice, an idiosyncratic productivity increase raises labor supply, holding aggregate shadow prices fixed. The consumption share is unchanged.

> **Lecture addition: Intuition for Complete Markets**
>
> - **Example**: 18 students trade state-contingent claims before earnings risk is realized. Their incomes can differ while consumption shares remain fixed.
> - **Marx analogy**: "From each according to ability, to each according to needs." The planner equalizes weighted marginal utilities; individual income does not determine consumption.
> - **Empirical test**: With common preferences and fixed Pareto weights, idiosyncratic earnings shocks should not change consumption shares after controlling for aggregate conditions. Excess sensitivity motivates incomplete-market models.
>
> Recording: 00:13:54, 00:16:04

1.2. How to model incomplete markets?

1. **Exogenous**: assume some markets absent.
   1. e.g. save/borrow via noncontingent bonds; no state-contingent claims.
2. **Endogenous**: derive market incompleteness from frictions.
   1. e.g. limited commitment (participation constraints), private information.
   2. deeper but harder to do

Lecture addition: Course Roadmap

- **Methods**: Dynamic programming, numerical methods.
- **Background**: Equilibrium and welfare theorems.
- **Growth**: Basic models of economic growth.
- **Incomplete markets**: Endogenous (participation constraints/limited commitment), exogenous (asset structure), search models.

Recording: 00:31:25, 00:33:02

Lecture addition: Frictions for Endogenous Incompleteness

- **Limited commitment**: Contracts unenforceable. E.g., US car insurance limits; agents flee jurisdiction (e.g., to Frankfurt) to avoid liability → limits risk trading.
- **Private information**: Effort/talent unobservable. Income sharing → moral hazard (effort reduction) → complete insurance inefficient/impossible.

Recording: 00:22:10, 00:27:38

1.3. Planning problem

Set $n_t = A = 1$; $f(k) = F(k, 1)$. Given $k_0 \geq 0$. Discount factor $0 < \beta < 1$; depreciation $0 \leq \delta \leq 1$.

Choose sequences $c_t, k_{t+1} \geq 0$:

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t) \text{ s.t. } c_t + k_{t+1} = f(k_t) + (1-\delta)k_t. \tag{1.3}$$

- **Lagrangian**; current-value multipliers λ_t:

$$\mathscr{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t (f(k_t) + (1-\delta)k_t - c_t - k_{t+1})]. \tag{1.4}$$

- **Interior FOCs**:

$$\begin{aligned} c_t &: u'(c_t) = \lambda_t, \\ k_{t+1} &: \lambda_t = \beta \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta]. \end{aligned} \tag{1.5}$$

  Euler equation:

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta]. \tag{1.6}$$

- **Transversality condition**:

$$\lim_{T \to \infty} \beta^T \lambda_T k_{T+1} = \lim_{T \to \infty} \beta^T u'(c_T) k_{T+1} = 0. \tag{1.7}$$

  Discounted shadow value of terminal capital → 0.
- Increasing, concave u; concave f; finite discounted utility: feasibility + FOCs + TVC → global optimum. At corners, use KKT conditions.

1.4. Towards a recursive formulation

- Sequence approach: infinitely many dated choices + optimality conditions.
- Repeated trade-off: consume today vs. save for tomorrow.
- Stationary problem; current capital k_t summarizes relevant history.
- Recursion: continuation value summarizes future choices; same decision problem at every date.

Value function: maximal lifetime utility from initial capital k_0:

$$v^*(k_0) = \max_{\{c_t, k_{t+1}\}_{t\geq 0}} \sum_{t=0}^{\infty} \beta^t u(c_t). \tag{1.8}$$

Feasible sequences: $c_t, k_{t+1} \geq 0$; $c_t + k_{t+1} = f(k_t) + (1-\delta)k_t$, all $t \geq 0$.

Split date 0 from continuation:

$$v^*(k_0) = \max_{c_0, k_1} \left[u(c_0) + \beta \max_{\{c_t, k_{t+1}\}_{t\geq 1}} \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) \right]. \tag{1.9}$$

- Outer max: $c_0, k_1 \geq 0$; $c_0 + k_1 = f(k_0) + (1-\delta)k_0$.
- Inner max: given chosen k_1, same feasibility constraints for all $t \geq 1$. Discounting restarts at date 1 → inner value $v^*(k_1)$.

Continuation value absorbs future choices. **Bellman equation** (current capital k, next-period capital k'):

$$v^*(k) = \max_{c, k'} [u(c) + \beta v^*(k')] \tag{1.10}$$

subject to

$$c + k' = f(k) + (1-\delta)k, c, k' \geq 0. \tag{1.11}$$

Infinite sequence → one-period choice + discounted continuation value. Stationarity → same problem at every date.

Bellman equation = **functional equation**; unknown = function $v(k)$.

Questions:
- **Existence**: conditions for solution v?
- **Uniqueness**: conditions for unique v within chosen function class?
- **Equivalence**: when does $v = v^*$, with maximizing policy solving original sequence problem?

Answer:
- *Recursive Macroeconomic Theory* (RMT), "Dynamic Programming" (Ljungqvist & Sargent, 2018, ch. 3).

Lecture addition: Computational Advantage of DP

- **Lagrangian**: Euler conditions give a difference equation. Shooting starts from a guessed initial control and checks the terminal condition.
- **Recursive**: Bellman eq. → direct computation of **global solution** via iteration.
- **Convergence**: Under the contraction assumptions, VFI converges to a unique fixed point. A numerical discretization still introduces approximation error.

Recording: 01:06:37, 01:07:39

1.5. Dynamic programming

Bellman operator T: candidate continuation value $v \to$ updated value Tv.

$$(Tv)(k) = \max_{c,k'}[u(c) + \beta v(k')] \tag{1.12}$$

subject to

$$c + k' = f(k) + (1-\delta)k, c, k' \geq 0. \tag{1.13}$$

Bellman solution = **fixed point**:

$$Tv = v \Leftrightarrow (Tv)(k) = v(k) \text{ for all feasible } k. \tag{1.14}$$

Fixed-point theorems give existence/uniqueness conditions within chosen function space; assumptions matter.

1.6. Contraction mapping theorem

Theorem 1.6.1 (Contraction mapping theorem): Let (S, d) be a nonempty complete metric space and $T : S \to S$ a contraction: there exists $\beta \in (0, 1)$ such that

$$d(Tx, Ty) \leq \beta d(x, y) \text{ for all } x, y \in S. \tag{1.15}$$

Then T has a unique fixed point $v \in S$. For every $v_0 \in S$ and $n = 0, 1, 2, \ldots$,

$$d(T^n v_0, v) \leq \beta^n d(v_0, v). \tag{1.16}$$

Hence $T^n v_0 \to v$ as $n \to \infty$, with geometric convergence. Here T^n denotes n successive applications of T, with T^0 the identity.

Lecture addition: Intuition for Contraction Mapping

- On a compact interval, $|g'(x)| \leq \beta < 1$ makes g a contraction, provided it maps the interval into itself. Brouwer gives existence for a continuous self-map; contraction also gives uniqueness and convergence.
- In function spaces, "slope" = contraction factor β. Iterating operator moves points closer → convergence to unique fixed point.

Recording: 00:59:06, 01:01:46

1.6.1. Application to planner

- Assume feasible capital remains in nonempty compact interval K.
- $S = C(K)$: continuous real-valued functions on K; automatically bounded.
- **Sup-norm metric**:

$$d(v, w) = \|v - w\|_\infty = \sup_{k \in K} |v(k) - w(k)|. \tag{1.17}$$

  (S, d) complete.
- Must verify $T : S \to S$: updated values remain bounded and continuous. Then Bellman discounting gives

$$\|Tv - Tw\|_\infty \leq \beta \|v - w\|_\infty. \tag{1.18}$$

Theorem 1.6.1 → unique fixed point in S; value iteration converges from any $v_0 \in S$.

Lecture addition: Economic Interpretation of VFI

- $v^0 = 0$ is a zero terminal payoff.
- Tv^0 solves a one-period problem: consume all current resources.
- $T^2 v^0$ solves a two-period problem. Each application of T adds one period; $T^n 0$ is the n-period value with zero terminal utility.

Recording: 00:56:48, 00:58:31

1.7. Using the value function analytically

Assume u, f, and v differentiable; optimal choices interior. Substitute $c = f(k) + (1 - \delta)k - k'$ into Bellman:

$$v(k) = \max_{k'} [u(f(k) + (1 - \delta)k - k') + \beta v(k')], \tag{1.19}$$

over feasible k'.

- **FOC for k'**:

$$-u'(c) + \beta v'(k') = 0 \Rightarrow u'(c) = \beta v'(k'). \tag{1.20}$$

- **Envelope condition**: differentiate w.r.t. k, holding optimal k' fixed:

$$v'(k) = u'(c)[f'(k) + 1 - \delta]. \tag{1.21}$$

- **Combine**: apply envelope condition next period → usual Euler equation:

$$u'(c_t) = \beta v'(k_{t+1}) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]. \tag{1.22}$$

Same condition as sequence problem. At corners, use KKT conditions.

1.8. Using the value function numerically

Need numerical representation of $v(k)$.

- Simplest approach: restrict capital to finite grid

$$K_N = \left\{k^1, k^2, ..., k^N\right\} \subset K. \tag{1.23}$$

- Approximate value function by vector

$$\boldsymbol{v} = (v_1, v_2, ..., v_N)^\top \in \mathbb{R}^N, v_i \approx v\left(k^i\right). \tag{1.24}$$

One entry per capital grid point.

Algorithm 1.8.1 (Value function iteration (VFI)):

1. **Choose grid** $K_N = \left\{k^1, ..., k^N\right\}$. For each pair (i, j), compute

$$c_{ij} = f\left(k^i\right) + (1 - \delta)k^i - k^j. \tag{1.25}$$

Let J_i contain choices j with $c_{ij} \geq 0$ and finite $u\left(c_{ij}\right)$. Each J_i must be nonempty.

2. **Initial guess**: any $\boldsymbol{v}^0 \in \mathbb{R}^N$, e.g. all zeros. Set iteration $m = 0$ and tolerance $\varepsilon > 0$.
3. **Bellman update**: for every $i = 1, ..., N$,

$$v_i^{m+1} = \max_{j \in J_i}\left[u\left(c_{ij}\right) + \beta v_j^m\right]. \tag{1.26}$$

Use the old vector for all updates: $\boldsymbol{v}^{m+1} = T_N \boldsymbol{v}^m$.

4. **Convergence check**:

$$\Delta_m = \max_{1 \leq i \leq N}\left|v_i^{m+1} - v_i^m\right|. \tag{1.27}$$

If $\Delta_m \leq \varepsilon$, stop and return $\hat{\boldsymbol{v}} = \boldsymbol{v}^{m+1}$. Otherwise set $m \leftarrow m + 1$ and repeat step 3.

5. **Recover approximate policy**: for each i, choose

$$j_i \in \arg\max_{j \in J_i}\left[u\left(c_{ij}\right) + \beta \hat{v}_j\right]. \tag{1.28}$$

Set $k'\left(k^i\right) = k^{j_i}$ and $c\left(k^i\right) = c_{ij_i}$.

Why it converges: T_N is a β-contraction in the sup norm. Theorem 1.6.1 → unique grid fixed point $\boldsymbol{v}_N^*$; Algorithm 1.8.1 converges from any initial vector. At stopping, value-error bound:

$$\|\hat{\boldsymbol{v}} - \boldsymbol{v}_N^*\|_\infty \leq \frac{\beta}{1 - \beta}\Delta_m \leq \frac{\beta}{1 - \beta}\varepsilon. \tag{1.29}$$

This controls iteration error; grid approximation error remains.

Lecture addition: Convergence in General Equilibrium

- Under its assumptions, CMT guarantees convergence of individual value functions.
- Heterogeneous GE: convergence of **distribution** of states (assets, income, human capital) required.
- Distribution = higher-dimensional object; may have multiple fixed points → more complex than single-agent VFI.

Recording: 01:14:16, 01:15:18

Lecture addition: Curse of Dimensionality

- Grid-based VFI suffers from **curse of dimensionality**: n state variables → M^n grid points (exponential growth).
- Large n (e.g., 50) → computationally intractable.
- Sparse grids reduce the number of nodes. Mean field methods couple individual decisions with distribution dynamics; equilibrium convergence still requires its own argument.

Recording: 01:15:51, 01:16:53

Lecture addition: Complexity vs. Reality

- Model complexity driven by **expectation side**: agents anticipate others' actions → complex fixed-point problems.
- Evidence: real people not "fully forward-looking" as standard models assume.
- **Responses**: (1) Develop computational tools for complex models; (2) Simplify models to match cognitive complexity (behavioral/limited rationality).

Recording: 01:17:25, 01:19:00

2. Lecture 2 Oct 5

2.1. Recall

2.1.1. Bellman equation for the standard growth model

$$v(k) = \max_{c,k'}[u(c) + \beta v(k')], u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \tag{2.1}$$

subject to

$$c + k' = (1-\delta)k + k^{\alpha}, c > 0, k' \geq 0. \tag{2.2}$$

For $\sigma = 1$, use $u(c) = \log c$.

2.1.2. Contraction mapping theorem

On a nonempty complete metric space (S, d), a contraction $T : S \to S$ with modulus $\beta \in (0, 1)$ has a unique fixed point v (Theorem 1.6.1):

$$\begin{aligned} d(Tx, Ty) &\leq \beta d(x, y), \\ d(T^n v_0, v) &\leq \beta^n d(v_0, v). \end{aligned} \tag{2.3}$$

Iteration from any $v_0 \in S$ converges geometrically to v.

2.2. Blackwell's sufficient conditions

Theorem 2.2.1 (Blackwell's sufficient conditions): Let $B(X)$ be the bounded real-valued functions on X, equipped with the sup norm, and let $T : B(X) \to B(X)$. Suppose:

1. **Monotonicity**: For all $f, g \in B(X)$,

$$f(x) \geq g(x) \text{ for all } x \in X \implies (Tf)(x) \geq (Tg)(x) \text{ for all } x \in X. \tag{2.4}$$

2. **Discounting**: There exists $\beta \in (0, 1)$ such that, for every $f \in B(X)$ and $a \geq 0$,

$$(T(f + a))(x) \leq (Tf)(x) + \beta a \text{ for all } x \in X. \tag{2.5}$$

   Here $(f + a)(x) = f(x) + a$.

Then T is a contraction with modulus β:

$$\|Tf - Tg\|_{\infty} \leq \beta \|f - g\|_{\infty}. \tag{2.6}$$

2.2.1. Verification for the Bellman operator

Maximization preserves pointwise inequalities, so $f \geq g \implies Tf \geq Tg$. Adding a constant to continuation utility gives

$$(T(f + a))(k) = \max_{c,k'}[u(c) + \beta f(k')] + \beta a = (Tf)(k) + \beta a. \tag{2.7}$$

Both conditions hold; the self-map requirement $T : B(X) \to B(X)$ must also be verified.

2.3. Blackwell's conditions for the standard model

Let $\alpha, \delta, \beta \in (0,1)$. Resources and consumption:

$$\begin{aligned} y(k) &= \underbrace{(1-\delta)k}_{\text{remaining capital}} + \underbrace{k^{\alpha}}_{\text{output}}, \\ c(k,k') &= y(k) - k'. \end{aligned} \tag{2.8}$$

The invariant state space satisfies

$$\begin{aligned} \bar{k} &= \delta^{-\frac{1}{1-\alpha}}, \\ X &= \left[0, \bar{k}\right], \\ 0 \le y(k) &\le y\left(\bar{k}\right) = \bar{k} \text{ for all } \; k \in X. \end{aligned} \tag{2.9}$$

> **Lecture addition: The capital ceiling**
>
> $\bar{k}$ is the steady state with zero consumption: all output goes into investment, and $\bar{k}^{\alpha} = \delta\bar{k}$. The bound comes from feasibility alone. Starting in X, feasible savings remain in X.
>
> Recording: 00:07:02 - 00:07:39

Feasible savings and the Bellman operator:

$$\begin{aligned} \Gamma(k) &= \{k' \ge 0 : c(k,k') \ge 0\} \subseteq X, \\ (Tf)(k) &= \sup_{k' \in \Gamma(k)} \left[\underbrace{u(c(k,k'))}_{\text{current utility}} + \beta \underbrace{f(k')}_{\text{continuation value}} \right]. \end{aligned} \tag{2.10}$$

Extend u to $c = 0$ by its limit, possibly $-\infty$. Replace sup with max when the optimum is attained.

2.3.1. Monotonicity

For $f, g \in B(X)$ with $f \ge g$ pointwise,

$$\begin{aligned} (Tf)(k) &= \sup_{k' \in \Gamma(k)} \left[u(c(k,k')) + \beta f(k')\right] \\ &\ge \sup_{k' \in \Gamma(k)} \left[u(c(k,k')) + \beta g(k')\right] \\ &= (Tg)(k). \end{aligned} \tag{2.11}$$

Lecture addition: Why the first inequality can be strict

If the maximum for g is attained, let $k^\star(k)$ be its optimal savings:

$$\begin{aligned}(Tf)(k) &\geq u(c(k,k^\star(k))) + \beta f(k^\star(k)),\\ &\geq u(c(k,k^\star(k))) + \beta g(k^\star(k)),\\ &= (Tg)(k).\end{aligned}$$

The first step evaluates one feasible choice. It can be strict because $k^\star(k)$ need not maximize the objective with continuation value f.

Recording: 00:08:00 - 00:10:59

2.3.2. Discounting

For $f \in B(X)$ and a constant $a \geq 0$,

$$\begin{aligned}(T(f+a))(k) &= \sup_{k' \in \Gamma(k)} \left[u(c(k,k')) + \beta f(k') + \underbrace{\beta a}_{\text{constant in } k'} \right] \\ &= (Tf)(k) + \beta a.\end{aligned} \tag{2.12}$$

2.3.3. Boundedness and conclusion

For $0 < \sigma < 1$, $u(0) = 0$ and

$$\begin{aligned}0 \leq u(c(k,k')) &\leq u\left(\overline{k}\right) < \infty,\\ \|Tf\|_\infty &\leq u\left(\overline{k}\right) + \beta \|f\|_\infty < \infty.\end{aligned} \tag{2.13}$$

Thus $T : B(X) \to B(X)$ on the complete sup-norm space $B(X)$. By Theorem 2.2.1 and Theorem 1.6.1, there is a unique $v \in B(X)$ with $Tv = v$, and for every $v_0 \in B(X)$,

$$\|T^n v_0 - v\|_\infty \leq \beta^n \|v_0 - v\|_\infty \to 0. \tag{2.14}$$

For $\sigma \geq 1$ (including log utility),

$$k = 0 \implies \Gamma(k) = \{0\} \implies (Tf)(0) = -\infty. \tag{2.15}$$

The bounded-function theorem then requires a different state or function space.

2.4. Formulating recursive models

Identify the state variables before writing the Bellman equation.

A **state variable**:
- Matters for the decision problem.
- Is taken as given by the decision maker at the time of choice.
- Varies over time or across states.

Other variables:
- **Choice variables**: chosen by the decision maker.

- **Parameters**: affect the problem but are fixed over time and across states.

The value function takes the state variables as its arguments.

> **Lecture addition: A past choice can be a current state**
>
> The household chose today's capital in the past; it is fixed when today's consumption-saving decision is made. Models with habits or varying patience may require preference states. Here β and σ remain parameters because they are fixed.
>
> Recording: 00:16:22 - 00:18:54

2.4.1. How to determine the state

1. **Fix timing**: Specify what is observed before choices are made. A lagged choice may be today's state; a newly chosen quantity is a control.
2. **List candidates**: Retain inherited stocks and observed shocks that affect current constraints, utility, or forecasts. Separate fixed parameters.
3. **Check sufficiency**: Given state s and choices q, specify

$$\begin{aligned} q &\in \Gamma(s), \\ \text{current utility} &= U(s,q), \\ s' &= G(s,q,\varepsilon'). \end{aligned} \tag{2.16}$$

   Here ε' is future uncertainty. Two histories with the same s must give the same feasible choices, current utility, and conditional distribution of s' for every feasible q.
4. **Remove redundancy**: Drop a coordinate only if this sufficiency test still holds. A one-to-one change of coordinates gives an equivalent state.
5. **Write the Bellman equation**:

$$v(s) = \sup_{q \in \Gamma(s)} \left[\underbrace{U(s,q)}_{\text{current utility}} + \beta \underbrace{\mathbb{E}[v(s') \mid s,q]}_{\text{expected continuation}} \right]. \tag{2.17}$$

   Use max when the optimum is attained; omit the expectation in a deterministic model.

Reduction test: If two candidate states have the same proposed summary but different feasible choices, utility, or transition distributions, the summary is insufficient.

2.4.2. Example: standard model with two sectors

Each period, allocate inherited capital k and one unit of labor freely between consumption (c) and investment (i) sectors. Let $\alpha, \gamma \in (0,1)$.

Feasibility:

$$
\begin{aligned}
c &= \underbrace{k_c^{\alpha} n_c^{1-\alpha}}_{\text{consumption output}}, \\
k' &= \underbrace{(1-\delta)k}_{\text{remaining capital}} + \underbrace{k_i^{\gamma} n_i^{1-\gamma}}_{\text{investment output}}, \\
k_c + k_i &= k, \\
n_c + n_i &= 1.
\end{aligned}
\tag{2.18}
$$

All quantities are nonnegative.

State identification:

- k is inherited; (k_c, k_i, n_c, n_i) are chosen before production.
- Free reallocation removes dependence on the previous sectoral split.
- Given k and current choices, utility and next-period k' are determined.

Hence $s = k$ is sufficient.

The Bellman equation is

$$
v(k) = \max_{c,k',k_c,k_i,n_c,n_i} [u(c) + \beta v(k')], \tag{2.19}
$$

subject to the constraints above, with u as defined in Recall.

> **Lecture addition: Timing and the sectoral split**
>
> Let $\tilde{v}(k_c, k_i)$ be the value after capital has been allocated across sectors. Before allocation,
>
> $$v(k) = \max_{k_c + k_i = k} \tilde{v}(k_c, k_i).$$
>
> In this deterministic model, the future split can be planned a period earlier. Treating the initial allocation consistently gives the same choices. With M grid points per state, retaining both stocks requires M^2 combinations; retaining total capital requires M.
>
> Recording: 00:24:30 - 00:30:44

2.4.3. Example: two sectors with irreversible investment

Capital is installed before the period and cannot move between sectors. Choose labor allocations and new investment x_c, x_i today; investment becomes productive next period.

Feasibility:

$$\begin{aligned} c &= k_c^{\alpha} n_c^{1-\alpha}, \\ x_c + x_i &= \underbrace{k_i^{\gamma} n_i^{1-\gamma}}_{\text{investment output}}, \\ k_c' &= (1-\delta)k_c + x_c, \\ k_i' &= (1-\delta)k_i + x_i, \\ n_c + n_i &= 1, \\ x_c, x_i &\geq 0. \end{aligned} \tag{2.20}$$

All other quantities are nonnegative. Irreversibility implies

$$k_j' \geq (1-\delta)k_j \text{ for } j \in \{c, i\}. \tag{2.21}$$

Sectoral capital can decline only through depreciation.

State: (k_c, k_i). Two economies with the same total capital but opposite sectoral splits face different current production possibilities and different bounds on tomorrow's stocks. Total capital alone is insufficient. Current labor and investment allocations are controls.

The Bellman equation is

$$v(k_c, k_i) = \max_{c, n_c, n_i, x_c, x_i} \{u(c) + \beta v((1-\delta)k_c + x_c, (1-\delta)k_i + x_i)\}, \tag{2.22}$$

subject to the feasibility constraints above.

Lecture addition: Why the aggregate equation is insufficient

The state is (k_c, k_i). Adding the accumulation equations leaves the separate restrictions $x_c, x_i \geq 0$ in place. Dropping them would permit conversion of existing capital between sectors. They can bind when initial stocks are imbalanced or shocks change the desired sectoral mix.

Recording: 00:32:20 - 00:34:45

Lecture addition: Applications of multisector models

Separate consumption and investment sectors allow general and investment-specific technological change to have different effects on business cycles. Larger models use input-output networks, with sectors buying CES aggregates of intermediate inputs, to study shock propagation.

Recording: 00:20:42 - 00:20:53; 00:35:00 - 00:36:15

2.4.4. Standard model with shocks

Productivity z is i.i.d. over time. The resource constraint is

$$c + k' = (1-\delta)k + zk^{\alpha}. \tag{2.23}$$

In all formulations, $c, k' \geq 0$.

2.4.4.1. States (k, z): after the shock

Observe z, then choose c and k':

$$v(k, z) = \max_{c,k'} [u(c) + \beta \mathbb{E}_{z'}[v(k', z')]], \tag{2.24}$$

subject to Equation (2.23). The expectation is over next period's shock z'.

2.4.4.2. State x: total resources

After observing z, define

$$x = (1 - \delta)k + zk^{\alpha}. \tag{2.25}$$

Then

$$w(x) = \max_{c,k'} [u(c) + \beta \mathbb{E}_{z'}[w(x')]], \tag{2.26}$$

subject to

$$\begin{aligned} c + k' &= x, \\ x' &= (1 - \delta)k' + z'(k')^{\alpha}. \end{aligned} \tag{2.27}$$

z' is independent of z, so the transition of x' requires only k'.

Note: This reduction to x relies on z being i.i.d. over time.

> **Lecture addition: The forecast matters with persistent shocks**
>
> Two states can have the same current resources x but different productivity z. With persistence, their distributions of z' differ, so their optimal savings can differ. Current z carries information about tomorrow.
>
> Recording: 00:45:42 - 00:46:22

2.4.4.3. State k: before the shock

$V(k)$ is the value *before z is observed.* Choices $c(z)$ and $k'(z)$ may *depend on its realization:*

Choices. *For a fixed (k, z) or x, Options 1 and 2 maximize over scalar choices c, k'. Their policy functions are $c(k, z), k'(k, z)$ or $c(x), k'(x)$; state arguments are implicit. Option 3 maximizes over functions $c(z), k'(z)$ for every possible z.*

$$V(k) = \max_{c(z),k'(z)} \mathbb{E}_z[u(c(z)) + \beta V(k'(z))], \tag{2.28}$$

subject to

$$c(z) + k'(z) = (1 - \delta)k + zk^{\alpha} \text{ for every } z. \tag{2.29}$$

$$\begin{aligned} v(k, z) &= w((1 - \delta)k + zk^{\alpha}), \\ V(k) &= \mathbb{E}_z[v(k, z)]. \end{aligned} \tag{2.30}$$

Timing. Before observing z, $V(k)$ averages over its realization; i.i.d. shocks make past productivity uninformative about z. After observing z, the state records realized resources through (k, z) or x. Decisions can depend on observed z in all three formulations.

Lecture addition: Several values within one period

A model can use value functions at several stages of a period. A value before information arrives is linked to the later value by an expectation over the information revealed between those stages.

Recording: 00:51:25 - 00:52:16

2.5. Equilibrium

A social planner maximizes welfare subject to feasibility constraints. In a decentralized economy, agents make their own choices, linked through markets, institutions, and policies.

- How can decentralized outcomes be represented formally?
- How do equilibrium allocations relate to the planner's solution?
- Can decentralized equilibria be formulated recursively?

2.6. What is a competitive equilibrium?

A competitive equilibrium consists of:
- An *allocation*: consumption, production, and factor inputs.
- A *price system*: prices in every market.

Two conditions:
- **Optimization**: households maximize utility within their budgets; firms maximize profits. Both take prices as given. Household wealth includes endowments and ownership shares of firm profits.
- **Market clearing**: demand equals endowments plus net production for every good. With free disposal, excess supply is permitted only at a zero price.

Lecture addition: Trading dates and events

With complete markets, goods are distinguished by delivery date and event. The lecture's example is consumption in 2027 conditional on no rain in Australia. Suitable sequential asset markets can span the same contingent consumption trades as markets opened once at date 0.

Recording: 01:02:26 - 01:03:56

2.7. Sequential equilibrium

In the deterministic standard model, markets open each period. The household owns capital k_t and a fixed labor endowment $n_t = 1$. It rents capital and supplies labor to the firm; both agents take prices as given. Let r_t be the net return on capital. The capital rental rate is $r_t + \delta$; the wage is w_t.

Given k_0, a sequential equilibrium consists of an allocation $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ and prices $\{r_t, w_t\}_{t=0}^{\infty}$ satisfying the following conditions.

2.7.1. Household

Given the price paths, the household solves

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t), \tag{2.31}$$

subject to

$$\begin{aligned} c_t + k_{t+1} &= \underbrace{(1 + r_t)k_t}_{\text{capital payoff}} + \underbrace{w_t n_t}_{\text{labor income}}, \\ n_t &= 1, \\ c_t &\geq 0, \\ k_{t+1} &\geq 0. \end{aligned} \tag{2.32}$$

2.7.2. Firm

The firm chooses capital demand K_t and labor demand N_t each period:

$$\max_{K_t, N_t \geq 0} \left[K_t^{\alpha} N_t^{1-\alpha} - (r_t + \delta)K_t - w_t N_t \right]. \tag{2.33}$$

Lecture addition: Representative agents and price taking

The representative firm behaves as an atomistic price taker. Its labor demand N_t is a choice; $N_t = 1$ is imposed through market clearing. Representative notation can stand for a continuum of identical agents. Separate demand and supply symbols make this distinction explicit.

Recording: 01:06:36 - 01:08:06; 01:12:24 - 01:13:53

2.7.3. Market clearing

$$\begin{aligned} K_t &= k_t, \\ N_t &= n_t = 1, \\ c_t + k_{t+1} &= (1 - \delta)k_t + k_t^{\alpha}. \end{aligned} \tag{2.34}$$

At positive inputs, the firm's first-order conditions imply

$$\begin{aligned} r_t &= \alpha k_t^{\alpha - 1} - \delta, \\ w_t &= (1 - \alpha)k_t^{\alpha}. \end{aligned} \tag{2.35}$$

Lecture addition: Depreciation and the quoted return

An alternative convention quotes the capital rental rate $\rho_t = r_t + \delta$ and lets the household bear depreciation. Then

$$c_t + k_{t+1} = (1 - \delta + \rho_t)k_t + w_t n_t,$$
$$\text{firm capital cost} = \rho_t K_t.$$

The net-return and rental-rate conventions describe the same payments.

Recording: 01:10:41 - 01:11:28; 01:14:52 - 01:16:21

2.7.4. No-Ponzi condition

If borrowing is allowed, replace $k_{t+1} \geq 0$ with

$$\liminf_{t\to\infty} \frac{k_{t+1}}{\prod_{s=1}^{t}(1 + r_s)} \geq 0. \tag{2.36}$$

At an interior optimum with $u'(c_s) > 0$, the Euler equation gives

$$1 + r_s = \frac{u'(c_{s-1})}{\beta u'(c_s)}. \tag{2.37}$$

Multiplying from $s = 1$ to t, marginal utilities cancel:

$$\begin{aligned} \prod_{s=1}^{t}(1 + r_s) &= \prod_{s=1}^{t} \frac{u'(c_{s-1})}{\beta u'(c_s)}, \\ &= \frac{u'(c_0)}{\beta^t u'(c_t)}. \end{aligned} \tag{2.38}$$

Hence

$$\frac{k_{t+1}}{\prod_{s=1}^{t}(1 + r_s)} = \frac{\beta^t u'(c_t)k_{t+1}}{u'(c_0)}. \tag{2.39}$$

Since $u'(c_0) > 0$ is constant, dividing by it preserves the sign of the limit inferior. Thus the no-Ponzi condition is equivalent to

$$\liminf_{t\to\infty} \beta^t u'(c_t)k_{t+1} \geq 0. \tag{2.40}$$

The household cannot finance consumption by rolling over debt forever.

Lecture addition: Physical capital and financial assets

Physical capital is nonnegative. An individual's financial asset position can be negative when borrowing is allowed for consumption smoothing. Period budgets alone permit arbitrarily large debt followed by refinancing of principal and interest. A borrowing limit or no-Ponzi condition restricts those plans.

Recording: 01:16:28 - 01:17:20; 01:18:29 - 01:19:47

2.8. Arrow-Debreu equilibrium

Sequential markets open each period. In an Arrow-Debreu equilibrium, markets open once at date 0. Agents contract for goods delivered at each future date (and state, under uncertainty). Delivery and production still occur each period.

Given k_0, an Arrow-Debreu equilibrium consists of an allocation $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ and prices $\{p_t, r_t, w_t\}_{t=0}^{\infty}$ satisfying the conditions below.

The additional price $p_t > 0$ is the date-0 price of one unit of the good delivered at date t, with $p_0 = 1$. It converts date-t quantities into date-0 goods.

The wage w_t and capital rental rate $r_t + \delta$ remain measured in date-t goods. Their date-0 prices are $p_t w_t$ and $p_t(r_t + \delta)$.

2.8.1. Household

The household solves

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t), \tag{2.41}$$

subject to $n_t = 1$, $c_t \geq 0$, $k_{t+1} \geq 0$, given k_0, and a single present-value budget:

$$\sum_{t=0}^{\infty} p_t(c_t + k_{t+1}) = \sum_{t=0}^{\infty} p_t[(1 + r_t)k_t + w_t n_t]. \tag{2.42}$$

The present-value sums are assumed finite.

2.8.2. Firm and market clearing

For each delivery date t, the firm solves

$$\max_{K_t, N_t \geq 0} \left[K_t^{\alpha} N_t^{1-\alpha} - (r_t + \delta)K_t - w_t N_t\right]. \tag{2.43}$$

Multiplying profits by $p_t > 0$ leaves the input choices unchanged. Markets clear at every date:

$$\begin{aligned} K_t &= k_t, \\ N_t &= n_t = 1, \\ c_t + k_{t+1} &= (1 - \delta)k_t + k_t^{\alpha}. \end{aligned} \tag{2.44}$$

2.8.3. No arbitrage and the present-value budget

Definition 2.8.1 (No arbitrage in capital): In this deterministic Arrow-Debreu economy, one unit of k_{t+1} costs p_t at date 0 and pays $p_{t+1}(1 + r_{t+1})$ in date-0 goods. At positive capital holdings, no arbitrage requires

$$\begin{aligned} p_t - p_{t+1}(1 + r_{t+1}) &= 0, \\ \frac{p_t}{p_{t+1}} &= 1 + r_{t+1}. \end{aligned} \tag{2.45}$$

Thus $\frac{p_t}{p_{t+1}}$ is the gross real return from t to $t + 1$.

Lecture addition: Why positive capital requires no arbitrage

One extra unit of k_{t+1} changes lifetime wealth by

$$p_{t+1}(1 + r_{t+1}) - p_t.$$

- If negative, the household chooses $k_{t+1} = 0$.
- If positive, capital purchases generate unbounded wealth; there is no finite optimum.
- If zero, the household is indifferent over capital holdings at these prices.

An equilibrium with positive capital therefore requires equality. Future capital purchases and payoffs cancel; **initial capital** remains part of wealth because its acquisition cost is outside the lifetime budget.

Recording: 2026-10-06, 00:00:46 - 00:03:31

With $p_0 = 1$,

$$p_t = \frac{1}{\prod_{s=1}^{t}(1 + r_s)}. \tag{2.46}$$

Capital purchases and their future payoffs cancel from the lifetime budget:

$$\sum_{t=0}^{\infty} p_t c_t = \underbrace{(1 + r_0)k_0}_{\text{initial capital wealth}} + \underbrace{\sum_{t=0}^{\infty} p_t w_t}_{\text{lifetime labor income}}. \tag{2.47}$$

The lifetime budget makes a separate no-Ponzi condition unnecessary.

Lecture addition: Valuing other assets

Date prices also value a claim to a dividend stream d_t:

$$q_0 = \sum_{t=1}^{\infty} p_t d_t.$$

The same method prices a bond by multiplying each payoff by its delivery-date price. With uncertainty, use **state-contingent prices**. This is why Arrow-Debreu notation is convenient for asset pricing.

Recording: 2026-10-06, 00:04:08 - 00:05:13

2.9. Recursive equilibrium

Recursive methods apply to **decentralized economies** as well as planning problems. Build on **sequential equilibrium**: agents choose each period, and state variables evolve over time.

In the standard model, distinguish two states:
- k: **individual capital**, which determines household wealth.
- K: **aggregate capital**, which determines prices and their future path.

The household chooses k' taking $K' = g(K)$ and the associated prices *as given*. In equilibrium, $k = K$, but keeping both states in the household problem preserves **price-taking**. With heterogeneity, individual and aggregate states generally differ.

Lecture addition: Whose saving moves tomorrow's prices?

Think of the representative household as one of many identical, tiny households. More **aggregate** saving lowers tomorrow's return to capital and raises wages. One household's saving has no effect on either price.

Thus $K' = g(K)$ is taken as given inside the household problem. Impose $g(K) = k'(K, K)$ when checking equilibrium consistency.

Recording: 2026-10-06, 00:07:30 - 00:08:48; 00:11:18 - 00:11:46

2.9.1. Standard model

A recursive competitive equilibrium consists of:
- **Value function:** $v(k, K)$.
- **Policy functions:** $c(k, K)$ and $k'(k, K)$.
- **Price functions:** $r(K)$ and $w(K)$.
- **Aggregate law of motion:** $g(K)$.

These functions satisfy four conditions:

1. **Household optimality.** Given r, w and g, the value function solves

$$v(k, K) = \max_{c, k' \geq 0} \{u(c) + \beta v(k', K')\}, \tag{2.48}$$

subject to

$$\begin{aligned} c + k' &= (1 + r(K))k + w(K), \\ K' &= g(K). \end{aligned} \tag{2.49}$$

The policies $c(k, K)$ and $k'(k, K)$ attain this maximum.

2. **Firm optimality.** With aggregate labor equal to one and $K > 0$,

$$\begin{aligned} r(K) &= \alpha K^{\alpha-1} - \delta, \\ w(K) &= (1 - \alpha)K^{\alpha}. \end{aligned} \tag{2.50}$$

Here $r(K)$ is the *net return*; the capital rental rate is $r(K) + \delta$.

3. **Market clearing.** For every admissible K,

$$c(K, K) + k'(K, K) = (1 - \delta)K + K^{\alpha}. \tag{2.51}$$

4. **Consistency.** The *perceived* aggregate law must equal the law *generated* by optimal household saving:

$$g(K) = k'(K, K). \tag{2.52}$$

2.10. The First Welfare Theorem

Theorem 2.10.1 (First Welfare Theorem): With **locally nonsatiated preferences**, **complete competitive markets** and **no externalities**, every competitive equilibrium allocation is **Pareto efficient**. This formulation assumes finitely many consumers and well-defined lifetime budgets.

Pareto efficiency means that no feasible allocation makes everyone weakly better off and at least one person strictly better off.

2.10.1. Key assumptions

- **Local nonsatiation:** every neighborhood of a consumption bundle contains a strictly preferred bundle.
- **Complete markets:** all relevant goods, delivery dates and states can be traded.
- **No externalities:** effects on others operate through market prices and trades.

Convexity is not required. Economies with *infinitely many overlapping generations* lie outside this finite-consumer result; competitive equilibria can be inefficient.

Lecture addition: Why overlapping generations are an exception

In a two-period OLG economy, infinitely many generations can trade an intrinsically worthless asset. The young give goods to the old for money, then sell that money to the next young generation when they retire. Under suitable conditions, this transfer scheme can improve on a nonmonetary equilibrium.

A last generation breaks the argument: it cannot sell the money onward and loses from accepting it. The infinite sequence of consumers matters for this exception to the First Welfare Theorem.

Recording: 2026-10-06, 00:18:10 - 00:21:32

2.10.2. Proof by contradiction of Theorem 2.10.1

Proof: Let $p \geq 0$ denote equilibrium prices and ω aggregate endowments. At equilibrium, consumer i consumes $x_i^\star$ and has wealth m_i from endowments and distributed profits; firm j produces $y_j^\star$.

Suppose $(\tilde{x}_i, \tilde{y}_j)$ is a feasible Pareto improvement: every consumer weakly prefers $\tilde{x}_i$ to $x_i^\star$, and consumer h strictly prefers $\widetilde{x_h}$ to $x_h^\star$.

1. **Consumer optimality.**
   - **Strict improvement:** the bundle must cost more than m_i; otherwise it was affordable.
   - **Weak improvement:** the bundle cannot cost less than m_i. With *budget slack*, local nonsatiation gives a nearby, strictly better bundle that remains affordable.

   Either violation contradicts optimality. Hence

   $$\begin{aligned} &p \cdot \tilde{x}_i \geq m_i \text{ for every } i, \\ &p \cdot \widetilde{x_h} > m_h \text{ for the strictly improved consumer } h. \end{aligned} \tag{2.53}$$

   Summing gives

   $$\sum_i p \cdot \tilde{x}_i > \sum_i m_i. \tag{2.54}$$

2. **Feasibility and firm optimality.** Consumption is bounded by endowments plus production.

   Profit maximization implies $p \cdot \tilde{y}_j \leq p \cdot y_j^\star$ for each firm. Therefore

   $$\begin{aligned} \sum_i p \cdot \tilde{x}_i &\leq p \cdot \omega + \sum_j p \cdot \tilde{y}_j \text{ (feasibility)}, \\ &\leq p \cdot \omega + \sum_j p \cdot y_j^\star \text{ (profit maximization)}, \\ &= \sum_i m_i \text{ (endowments + distributed profits)}. \end{aligned} \tag{2.55}$$

3. **Contradiction.** Total cost cannot be both *greater than* and *at most* aggregate wealth. No feasible Pareto improvement exists.

□

2.10.3. Implication for the standard model

In the **representative-agent standard model**, a competitive equilibrium allocation solves the planner's problem. *If the planner's solution is unique and equilibrium exists*, the allocations coincide. Recover prices from firms' marginal products.

2.11. The Second Welfare Theorem

Theorem 2.11.1 (Second Welfare Theorem): With complete competitive markets, no externalities and the conditions below, any **Pareto-efficient allocation** can be supported as a **competitive equilibrium** after **lump-sum wealth transfers**.

2.11.1. Sufficient conditions

- **Consumers:** a finite set.
- **Consumption:** each consumption set X_i is convex.
- **Preferences:** u_i is continuous and locally nonsatiated. For $x, x' \in X_i$ and $0 < \lambda < 1$, assume

$$u_i(x) > u_i(x') \Longrightarrow u_i(\lambda x + (1-\lambda)x') > u_i(x'). \tag{2.56}$$

- **Production:** the aggregate production set $Y = \sum_j Y_j$ is convex.
- **Separation:** the commodity space S is finite-dimensional, or Y has a nonempty interior.
- **Budget regularity:** at supporting prices p, each consumer has a feasible bundle $x_i^0 \in X_i$ with

$$p \cdot x_i^0 < p \cdot x_i^\star. \tag{2.57}$$

Convexity alone guarantees only a *quasi-equilibrium*: a strictly preferred bundle may cost exactly the consumer's wealth.[1]

2.11.2. Wealth transfers

At supporting prices, m_i includes original endowments and profits at the target production plan.

For target consumption $x_i^\star$, choose **transfers** T_i:

$$\begin{aligned} T_i &= \underbrace{p \cdot x_i^\star}_{\text{target wealth}} - \underbrace{m_i}_{\text{original wealth}}, \\ p \cdot x_i &\le m_i + T_i = p \cdot x_i^\star, \\ \sum_i T_i &= 0. \end{aligned} \tag{2.58}$$

[1] See Condie's welfare-theorem notes, Proposition 6.1 ° for the budget condition. In an infinite-dimensional commodity space, $p \cdot x$ denotes the continuous price functional applied to x.

Feasibility and profit distribution imply $\sum_i p \cdot x_i^\star = \sum_i m_i$, so transfers sum to zero. **Lump-sum** transfers are *independent of choices*, leaving marginal incentives unchanged.

2.11.3. Supporting prices

1. **Separation:** convexity permits a separating hyperplane between feasible aggregate resources and aggregate bundles that improve welfare.
2. **Prices:** the separating functional defines the price system p.
3. **Optimization:** firms maximize profits at these prices; budget regularity ensures consumers optimize their transfer-adjusted budgets.

2.12. How do the welfare theorems matter?

- **Markets and information:** under the welfare-theorem assumptions, equilibrium prices coordinate decentralized choices into an efficient allocation. *The Use of Knowledge in Society* (Hayek, 1945) explains how prices communicate *dispersed information*.
- **Policy:** diagnose failed assumptions, such as externalities, market power or missing markets. Under the second theorem's conditions, *lump-sum redistribution* can implement different efficient distributions without changing marginal incentives.
- **Computing equilibrium:** in the standard model, solve the planner's Bellman equation, then recover prices from marginal products. With heterogeneity, a planner allocation may require *wealth transfers* to be decentralized.

Lecture addition: Diagnosing a policy problem

Ask which welfare-theorem assumption fails in the proposed application.

- **Pollution:** an upstream producer harms downstream users without paying for that harm. A market for the relevant rights can price the externality.
- **Health insurance:** adverse selection can prevent useful insurance markets from forming. A missing market may justify intervention, but the underlying information problem still needs to be addressed.

Recording: 2026-10-06, 00:27:40 - 00:29:30

Lecture addition: What should an economist remember?

The class proposed equilibrium, Nash equilibrium, supply and demand, and economy-wide resource constraints. **Nash equilibrium** concerns individual best responses; maximizing total welfare is a different condition.

Economics often starts from **purposeful choices**: agents have objectives and face constraints. Macroeconomics asks how these choices interact and add up. Giving one group more resources requires accounting for where those resources come from.

Recording: 2026-10-06, 00:32:07 - 00:39:19

3. Lecture 3 Oct 6

3.1. Fundamental models of economic growth

The **standard model** is a Solow growth model with *optimizing consumers*. It matches long-run growth facts for high-income countries, but growth models should also explain:

- Economic development beyond the period described by the Kaldor facts.
- Income and growth differences across *all countries*.
- The origin of productivity growth.

Definition 3.1.1 (Kaldor's growth facts): Kaldor's six **stylized facts** (Kaldor, 1961) summarize long-run growth patterns, originally observed in industrialized economies:

1. **Output per worker** $\frac{Y_t}{N_t}$ grows at a roughly constant positive rate.
2. **Capital per worker** $\frac{K_t}{N_t}$ grows over time.
3. The **return to capital** is roughly constant.
4. The **capital-output ratio** $\frac{K_t}{Y_t}$ is roughly constant.
5. **Labor and capital income shares** are roughly constant.
6. **Output-per-worker growth rates** differ substantially across countries.

3.2. Solow

3.2.1. Setup

Output combines capital K_t and effective labor $A_t N_t$:

$$Y_t = K_t^{\alpha}(A_t N_t)^{1-\alpha}, 0 < \alpha < 1. \tag{3.1}$$

Here N_t is population (and labor supply); A_t is **labor-augmenting productivity**. Both grow *exogenously*:

$$\begin{aligned} A_{t+1} &= (1+g)A_t, \\ N_{t+1} &= (1+n)N_t. \end{aligned} \tag{3.2}$$

Assume $g, n \geq 0$. A **fixed saving rate** $0 < s < 1$ determines investment:

$$\begin{aligned} K_{t+1} &= (1-\delta)K_t + sY_t, \\ C_t &= (1-s)Y_t, \end{aligned} \tag{3.3}$$

where $0 \leq \delta \leq 1$ is depreciation. In the *standard model*, households instead choose saving optimally.

3.2.2. Capital per effective worker

Normalize by effective labor:

$$
\begin{aligned}
k_t &= \frac{K_t}{A_t N_t}, \\
y_t &= \frac{Y_t}{A_t N_t} = k_t^{\alpha}, \\
c_t &= \frac{C_t}{A_t N_t} = (1-s)k_t^{\alpha}.
\end{aligned}
\tag{3.4}
$$

Divide capital accumulation by $A_t N_t$ and use $A_{t+1}N_{t+1} = (1+g)(1+n)A_t N_t$:

$$
\begin{aligned}
k_{t+1}(1+g)(1+n) &= (1-\delta)k_t + sk_t^{\alpha}, \\
k_{t+1} - k_t &= \frac{sk_t^{\alpha} - (\delta + g + n + gn)k_t}{(1+g)(1+n)}.
\end{aligned}
\tag{3.5}
$$

Effective capital rises when investment exceeds depreciation and the capital needed to keep up with population and productivity growth. The term gn is retained because time is *discrete*.

3.2.3. Balanced growth path

On a **balanced growth path (BGP)**, capital per effective worker is constant:

$$
\begin{aligned}
s\bar{k}^{\alpha} &= (\delta + g + n + gn)\bar{k}, \\
\bar{k} &= \left(\frac{s}{\delta + g + n + gn}\right)^{\frac{1}{1-\alpha}}.
\end{aligned}
\tag{3.6}
$$

This positive steady state requires $\delta + g + n + gn > 0$. For $k_0 > 0$, diminishing returns imply convergence: k_t rises below $\bar{k}$ and falls above it.

- **Aggregate quantities:** K_t, Y_t and C_t grow at rate $(1+g)(1+n) - 1$.
- **Per capita quantities:** $\frac{K_t}{N_t}$, $\frac{Y_t}{N_t}$ and $\frac{C_t}{N_t}$ grow at rate g.
- **Growth facts:** the capital-output ratio, factor income shares and the marginal product of capital are constant; real wages grow at rate g.

A higher saving rate raises the BGP **level** of income per capita, not its **growth rate** g. With $g > 0$, the BGP matches Kaldor's first five facts. Cross-country differences in long-run growth require different *exogenous* productivity growth rates g.

Lecture addition: Growth during the transition

Constant $k_t = \frac{K_t}{A_t N_t}$ does not mean constant K_t: on the BGP, aggregate capital grows with effective labor.

Cobb-Douglas **factor income shares** are constant even outside the BGP. Starting below $\overline{k}$, k_t rises, so

$$\frac{K_t}{Y_t} = k_t^{1-\alpha} \text{ rises,}$$

$$\text{MPK}_t = \alpha k_t^{\alpha-1} \text{ falls.}$$

Constant capital-output ratios and returns are properties of the BGP, rather than every point along the transition.

Recording: 2026-10-06, 00:46:30 - 00:46:49; 00:48:04 - 00:48:30

Lecture addition: What the fixed saving rate assumes

The Solow model imposes the consumption-saving choice through s. It is a **reduced-form behavioral assumption**. The standard model derives this choice from household optimization.

A steady ratio in effective units still permits growth in levels and transitional dynamics. The version here is deterministic because it contains no shocks.

Recording: 2026-10-06, 00:50:44 - 00:52:16

3.2.4. Limitations

- Persistent cross-country income gaps require *exogenous differences* in productivity, saving or population growth.
- Baseline calibrations can imply faster convergence than observed, even among high-income countries.
- Productivity growth is unexplained; the model also lacks a mechanism for the historical transition from long-run stagnation to sustained growth.

3.3. Malthus

3.3.1. Setup

The model describes a preindustrial economy with population N_t and **fixed land** $X > 0$:

$$Y_t = N_t^{\alpha}(A_t X)^{1-\alpha}, 0 < \alpha < 1. \tag{3.7}$$

Productivity grows exogenously at rate $g \geq 0$:

$$A_{t+1} = (1+g)A_t. \tag{3.8}$$

Population growth responds to income per capita:

$$\begin{aligned} \frac{N_{t+1}}{N_t} &= s\frac{Y_t}{N_t}, \\ N_{t+1} &= sY_t. \end{aligned} \tag{3.9}$$

Here $s > 0$ is a demographic parameter. Higher income raises fertility or reduces mortality. Population increases when $s\frac{Y_t}{N_t} > 1$.

> **Lecture addition: Two routes from income to population**
>
> Higher income can raise demand for children or improve survival through better nutrition, especially in infancy. Either route gives a positive income-population feedback.
>
> Unlike Solow's saving rate, this s is **not a fraction of output saved**; the model does not require $s < 1$. Its demographic interpretation differs even though it occupies the same place in the accumulation equation.
>
> Recording: 2026-10-06, 00:57:22 - 00:59:04; 01:00:08 - 01:00:14

3.3.2. Population per effective land unit

Normalize by effective land $A_t X$:

$$\begin{aligned} n_t &= \frac{N_t}{A_t X}, \\ \frac{Y_t}{A_t X} &= n_t^{\alpha}, \\ \frac{Y_t}{N_t} &= n_t^{\alpha-1}. \end{aligned} \tag{3.10}$$

Using $N_{t+1} = sY_t$ gives

$$\begin{aligned} n_{t+1}(1+g) &= sn_t^{\alpha}, \\ n_{t+1} &= \frac{s}{1+g} n_t^{\alpha}. \end{aligned} \tag{3.11}$$

Since $\alpha < 1$, income per person falls as population grows relative to effective land.

3.3.3. Balanced growth path

On the BGP, $n_t = \bar{n}$ is constant:

$$\begin{aligned} \bar{n}(1+g) &= s\bar{n}^{\alpha}, \\ \bar{n} &= \left(\frac{s}{1+g}\right)^{\frac{1}{1-\alpha}}. \end{aligned} \tag{3.12}$$

For $n_0 > 0$, n_t converges to $\bar{n}$: it rises below the steady state and falls above it.

BGP income per capita and population density are

$$\begin{aligned} \frac{Y_t}{N_t} &= \bar{n}^{\alpha-1} = \frac{1+g}{s}, \\ \frac{N_t}{X} &= A_t \bar{n}. \end{aligned} \tag{3.13}$$

- Population and aggregate output grow at rate g.
- **Income per capita is constant**, even with positive productivity growth.
- Holding s and g fixed, higher productivity levels imply greater population density and the same BGP income per capita.

A permanent increase in the *level* of productivity initially raises income per person. Higher income accelerates population growth. As population rises relative to land, diminishing returns bring income per person back toward $\frac{1+g}{s}$.

Lecture addition: Lower fertility raises levels, not long-run growth

Holding g fixed, a lower demographic parameter s raises BGP income per person, $\frac{1+g}{s}$, while per capita growth remains zero. The lecture used **later marriage** as an example: fewer childbearing years can lower fertility at a given income and support higher living standards within the Malthusian regime.

Recording: 2026-10-06, 01:07:11 - 01:09:19

Lecture addition: What if land expands?

Fixed land creates diminishing returns to population. If land instead grows exogenously, the same normalization uses $A_t X_t$. With $\frac{X_{t+1}}{X_t} = 1 + g_X$, effective land grows at

$$1 + g_{AX} = (1+g)(1+g_X).$$

Replacing g by g_{AX} gives BGP income per person $\frac{1+g_{AX}}{s}$. Frontier expansion can raise this **level** and sustain faster population growth; a constant expansion rate still leaves long-run income per person constant.

Recording: 2026-10-06, 01:09:24 - 01:11:28

The model captures long preindustrial periods of stagnant income per person.

3.3.4. Malthus versus Solow

Solow accumulates **capital**; Malthus accumulates **population**. On the BGP, the return to the accumulated factor is constant. With competitive factor pricing, Malthus implies

$$\begin{aligned} w_t &= \alpha \frac{Y_t}{N_t} = \alpha \frac{1+g}{s}, \\ r_t^X &= (1-\alpha)\frac{Y_t}{X}, \end{aligned} \tag{3.14}$$

where r_t^X is the rental rate of land. *Wages are constant and land rents grow at rate g.*

In Solow, the return to capital is constant and wages grow at rate g.

3.3.5. From Malthus to Solow

Malthus describes preindustrial stagnation; Solow describes sustained growth in modern economies. Neither model alone explains the transition.

Lecture addition: What must change to escape stagnation?

The lecture outlined three mechanisms for the next class:

- **Structural change:** move production from land-dependent agriculture toward industry and services, whose capital and labor inputs can expand.
- **Demographic change:** weaken the positive income-fertility relationship; richer modern economies generally have lower birth rates.
- **Productivity growth:** explain why it accelerates and how it interacts with the other mechanisms.

A student also raised the scale effect in endogenous growth: more people can produce more ideas. The lecturer noted that this is another mechanism studied in the literature.

Recording: 2026-10-06, 01:13:28 - 01:15:52

Malthus to Solow (Hansen & Prescott, 2002) combines both technologies in one economy. Firms choose which to operate as productivity grows.

3.3.5.1. Technologies and resources

Both technologies are available in every period and produce the **same good**:

$$\begin{aligned} Y_{M,t} &= A_{M,t} K_{M,t}^{\alpha_1} N_{M,t}^{\alpha_2} X^{1-\alpha_1-\alpha_2}, \\ Y_{S,t} &= A_{S,t} K_{S,t}^{\theta} N_{S,t}^{1-\theta}. \end{aligned} \tag{3.15}$$

The Malthus sector M uses land; the Solow sector S uses only capital and labor. Assume $\alpha_1, \alpha_2 > 0$, $\alpha_1 + \alpha_2 < 1$ and $0 < \theta < 1$. Sectoral **TFP** grows exogenously:

$$A_{j,t+1} = \gamma_j A_{j,t}, \quad j \in \{M, S\}. \tag{3.16}$$

Here $\gamma_M \geq 1$ and $\gamma_S > 1$ are gross growth factors. Competitive firms allocate capital and labor freely across sectors:

$$\begin{aligned} K_t &= K_{M,t} + K_{S,t}, \\ N_t &= N_{M,t} + N_{S,t}, \\ X &= 1. \end{aligned} \tag{3.17}$$

One period represents a generation, so capital depreciates fully ($\delta = 1$). Land does not depreciate. Aggregate feasibility is

$$C_t + K_{t+1} = Y_{M,t} + Y_{S,t}. \tag{3.18}$$

3.3.5.2. Households and population

Households live for two periods and work only when young. A household born at t chooses positive consumption and nonnegative capital k_{t+1} and land x_{t+1}. With $\beta > 0$, its objective is

$$\max \log c_{1,t} + \beta \log c_{2,t+1}, \tag{3.19}$$

subject to

$$\begin{aligned} c_{1,t} + k_{t+1} + q_t x_{t+1} &= w_t, \\ c_{2,t+1} &= r^K_{t+1} k_{t+1} + \left(r^X_{t+1} + q_{t+1}\right) x_{t+1}. \end{aligned} \tag{3.20}$$

Here q_t is the land price and r^K_t, r^X_t are capital and land rental rates. With full depreciation, r^K_{t+1} is the entire payoff from one unit of capital saved at t.

There are N_t young households at t. Population follows a **calibrated** function G:

$$N_{t+1} = G\left(c_{1,t}\right) N_t. \tag{3.21}$$

G rises with consumption at low living standards and falls at high living standards. Each young household supplies one unit of labor. Asset and goods markets clear when

$$\begin{aligned} K_{t+1} &= N_t k_{t+1}, \\ 1 &= N_t x_{t+1}, \\ C_t &= N_t c_{1,t} + N_{t-1} c_{2,t}. \end{aligned} \tag{3.22}$$

With interior asset holdings, household optimality gives

$$\begin{aligned} c_{1,t} &= \frac{w_t}{1+\beta}, \\ q_t r^K_{t+1} &= r^X_{t+1} + q_{t+1}, \\ K_{t+1} &= N_t \frac{\beta}{1+\beta} w_t - q_t. \end{aligned} \tag{3.23}$$

The second equation equates the returns on capital and land. The third subtracts land purchases from the young generation's total saving.

3.3.5.3. Solow sector entry

Normalize the output price to one. At given factor prices, the minimum cost of producing one unit with the Solow technology is

$$\frac{1}{A_{S,t}} \left(\frac{r^K_t}{\theta}\right)^{\theta} \left(\frac{w_t}{1-\theta}\right)^{1-\theta}. \tag{3.24}$$

To test entry, evaluate this cost at the **Malthus-only** factor prices $r^K_{M,t}$ and $w_{M,t}$. Entry is profitable when

$$A_{S,t} > \underbrace{\left(\frac{r^K_{M,t}}{\theta}\right)^{\theta}\left(\frac{w_{M,t}}{1-\theta}\right)^{1-\theta}}_{\text{entry threshold}}. \tag{3.25}$$

If $A_{S,0} = 1$, then $A_{S,t} = \gamma_S^t$; the slide's factor $\frac{1}{\gamma_S^t}$ is therefore $\frac{1}{A_{S,t}}$.

Once entry occurs, factor prices adjust. If both sectors operate, their marginal products of capital and labor must coincide:

$$\begin{aligned} r^K_t &= \alpha_1 \frac{Y_{M,t}}{K_{M,t}} = \theta \frac{Y_{S,t}}{K_{S,t}}, \\ w_t &= \alpha_2 \frac{Y_{M,t}}{N_{M,t}} = (1-\theta)\frac{Y_{S,t}}{N_{S,t}}. \end{aligned} \tag{3.26}$$

The Malthus sector remains active at every finite date. With fixed land, scaling its capital and labor down together makes their marginal products arbitrarily large, so allocating a small amount to this sector remains worthwhile.

3.3.5.4. Transition

1. Initially, only the Malthus technology operates. Population growth absorbs productivity gains, keeping living standards constant. Solow productivity continues to grow while that sector is unused.
2. On the Malthusian growth path, wages and capital rental rates are constant, so the entry threshold is fixed. Rising $A_{S,t}$ eventually crosses it, and firms allocate capital and labor to the Solow sector.
3. The Solow sector has constant returns in capital and labor, with no fixed land input. In the calibrated economy, its share expands and income per person grows. The Malthus sector's share tends to zero. Population growth eventually falls through the assumed function G.

3.3.5.5. Limitations

- Productivity growth remains exogenous.
- The demographic transition comes from the calibrated function G; fertility and mortality decisions are not derived from household choices.
- The two technologies produce the same good. Requiring a separate agricultural good for subsistence would change the mechanism.

4. Lecture 4 Oct 12

4.1. Consumption, heterogeneity and policy

Lecture addition: Consumption functions and consumption smoothing

In the simple Keynesian example, consumption responds mechanically to current income. An illustrative marginal propensity to consume of 0.8 gives

$$\text{multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{1 - 0.8} = 5.$$

An optimizing household instead considers lifetime resources:

- A permanent income gain supports a larger increase in consumption.
- A one-off windfall is spread across future periods through saving.
- An announced future income change can affect consumption today.

The saving-and-borrowing example assumes **incomplete insurance**. With complete markets, idiosyncratic windfalls would be shared through state-contingent claims.

Recording: 2026-10-12, 00:08:21 - 00:16:02

Lecture addition: Hall's consumption test

Consumption already incorporates information about expected future income. Hall's test asks whether other current information helps forecast its next change (Hall, 1978).

For the exact random-walk benchmark, assume an unconstrained interior choice, a constant gross return R, $\beta R = 1$ and quadratic utility $u'(c) = a - bc$, with $b > 0$. The Euler equation gives

$$\begin{aligned} u'(c_t) &= \beta R E_t[u'(c_{t+1})], \\ a - bc_t &= a - bE_t[c_{t+1}], \\ E_t[c_{t+1} - c_t] &= 0. \end{aligned}$$

Quadratic utility gives certainty equivalence. Precautionary saving can break the exact random-walk result.

Paper detail: Hall used aggregate US data. Current income did not help predict consumption once current consumption was controlled for, but stock prices did. The original empirical conclusion was therefore qualified.

Recording: 2026-10-12, 00:16:03 - 00:22:07

Lecture addition: From New Keynesian models to HANK

Woodford combines optimizing households with nominal price or wage rigidities (Woodford, 2003). Interest rates and expected future rates affect consumption through households' Euler equations.

Households differ in wealth, liquidity and debt exposure. A household facing mortgage repricing can react differently from a wealthy saver. **HANK** (Heterogeneous Agent New Keynesian) models incorporate these differences; the paper discussed was *Monetary Policy According to HANK* (Kaplan et al., 2018).

Recording: 2026-10-12, 00:24:00 - 00:31:09

Lecture addition: Debt deflation

With fixed nominal debt D, a fall in the price level raises its real value:

$$P' = \frac{P}{2} \quad \Rightarrow \quad \frac{D}{P'} = 2\frac{D}{P}.$$

Bankruptcies can disrupt production and access to finance. The lecturer suggested this could be an important channel behind severe downturns, possibly more important than a household demand shortfall. A related model of firm borrowing and collateral is *Credit Cycles* (Kiyotaki & Moore, 1997).

Recording: 2026-10-12, 00:32:24 - 00:34:35

Lecture addition: Ricardian equivalence as a benchmark

Hold government spending fixed. Debt finance postpones taxes. If households are unconstrained, forward looking and bear the future tax liabilities, changing the timing of taxes leaves lifetime wealth unchanged (Barro, 1974).

Borrowing constraints, finite horizons and limited foresight can break this neutrality. The benchmark helps identify which assumptions make the timing of taxes matter.

Recording: 2026-10-12, 00:34:45 - 00:37:03

4.2. Facts on growth and demographics

Figures from Lecture 5, slides 4–7.

- Income and population data: Maddison Project Database 2023 (Bolt & van Zanden, 2024).
- Fertility data: UN World Population Prospects (United Nations Population Division, 2024), via Our World in Data°.

4.2.1. Income per capita over the long run

Income remains low for centuries, then takes off. Takeoff dates differ across countries; modern income gaps are large.

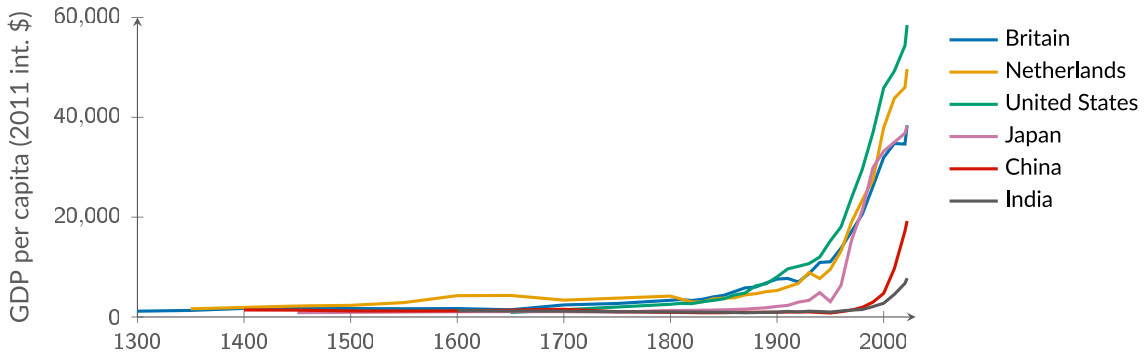


Figure 1: Income per capita over the long run.

Note: $y_t = \frac{Y_t}{N_t}$: income per capita in 2011 international dollars.

4.2.2. Population growth

Population growth has a hump shape in many regions, with peaks at different dates. Births, deaths and migration all matter.

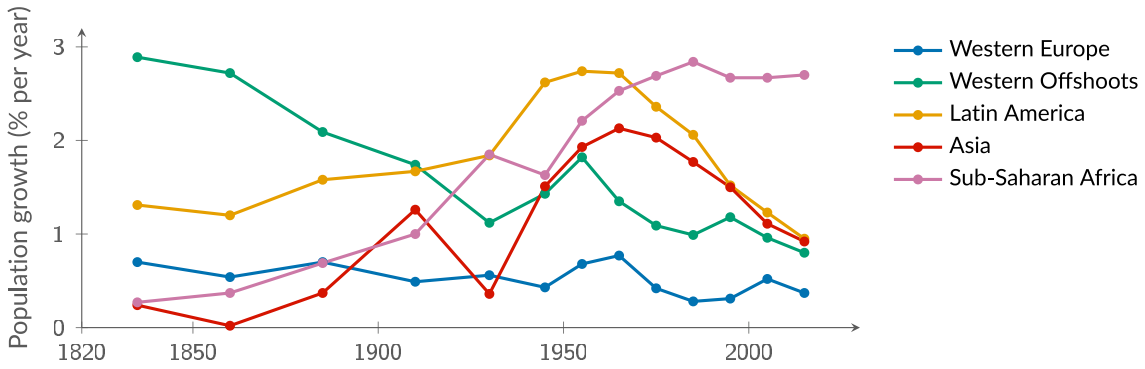


Figure 2: Population growth by region.

Note: g_N: average annual population growth (% per year) between benchmark years, plotted at interval midpoints.

4.2.3. Fertility over time

Total fertility rate (TFR) is lifetime births per woman at current age-specific fertility rates. Fertility declines at different times; several countries fall below replacement.

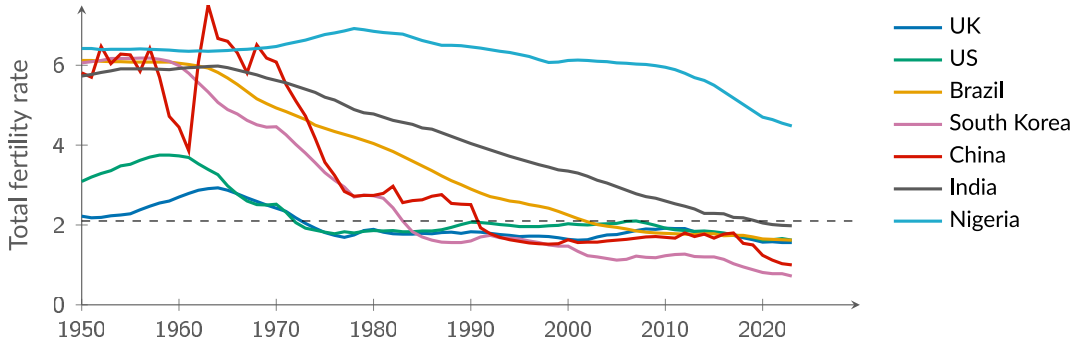


Figure 3: Fertility over time.

Note: TFR_t: total fertility rate (births per woman). Dashed line: replacement benchmark, about 2.1 births per woman.

4.2.4. Fertility and income across countries

In 2022, richer countries generally have lower fertility. Israel lies above the high-income cluster.

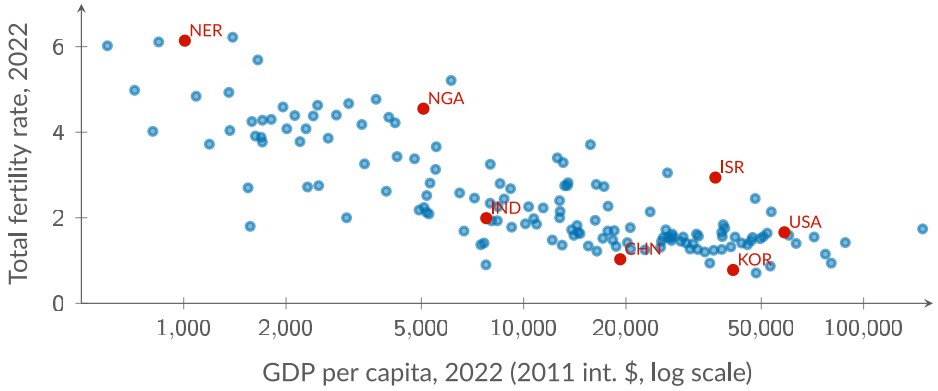


Figure 4: Fertility and income, 2022.

Note: TFR: births per woman; $y = \frac{Y}{N}$: income per capita in 2011 international dollars (log scale). Countries with population above one million.

4.3. Malthus to Solow: simulated transition

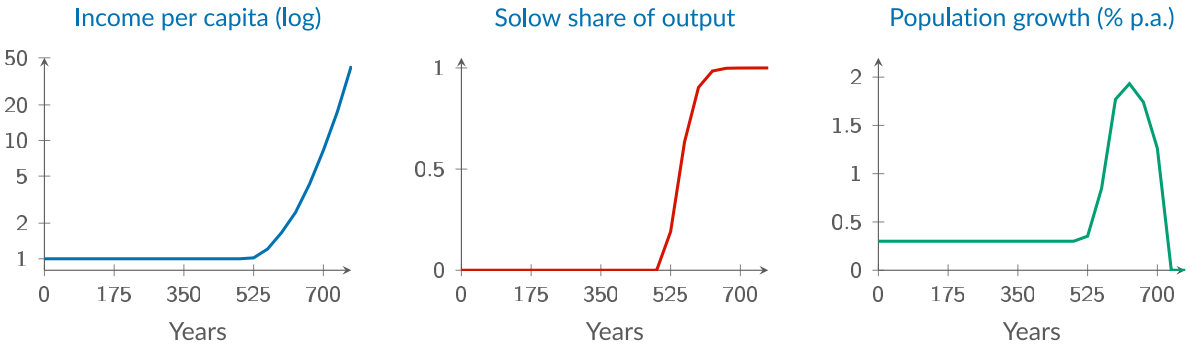


Figure 5: Simulated transition from Malthus to Solow.

Note: Left: income per capita $y = \frac{Y}{N}$ (model units, log scale). Middle: Solow output share $\frac{Y_S}{Y_M+Y_S}$. Right: population growth g_N (% per year). Time is in years; one model period is 35 years. Parameters: $(\alpha_1, \alpha_2, \theta) = (0.1, 0.6, 0.4)$; gross TFP growth factors per period: $(\gamma_M, \gamma_S) = (1.032, 1.518)$. Lecture 5, slide 12: simplified simulation using Hansen–Prescott technology parameters and population function (Hansen & Prescott, 2002), with a fixed saving rate in place of the OLG saving problem.

1. Initially, only the **Malthus sector** operates. Income per capita is flat; productivity gains raise population.
2. Rising Solow productivity crosses the **entry threshold**. The Solow sector's output share increases and income per capita starts to grow.
3. Population growth first rises, then falls. The decline comes from the **calibrated population function**; fertility choices are not derived here.

Lecture addition: Why an unused technology can improve

Solow-sector productivity grows even before the sector operates. The lecture interpreted this as accumulated knowledge from other activities: precise metalworking and other complementary skills eventually make industrial production viable. The steam engine illustrates how an invention can depend on many earlier improvements. Productivity growth is still exogenous in the model.

Recording: 2026-10-12, 00:54:30 - 00:56:12

Lecture addition: Trade and the return to education

A student asked how trade affects the transition. Galor and Mountford (Galor & Mountford, 2008) study specialization and demographic change:

- An early industrializer specializes in activities intensive in human capital and imports food. Higher returns to education favor investment per child.
- A later industrializer specializes in agriculture. Lower returns to education favor higher fertility and can delay its transition.

Trade can therefore widen differences in income per person and population growth through the quantity-quality choice.

Recording: 2026-10-12, 00:59:39 - 01:01:58

Lecture addition: Food and structural change

Hansen and Prescott treat the two sectors' outputs as perfect substitutes. If food must instead come from agriculture, a growing population may draw workers back into the land-dependent sector. Feeding the population then becomes a constraint on the expansion of industry.

Recording: 2026-10-12, 01:02:24 - 01:03:03

4.4. Becker, Murphy and Tamura (1990)

Human Capital, Fertility, and Economic Growth (Becker et al., 1990). Lecture 5, slides 14 to 28 use a simplified version without physical capital.

Parents choose the number of children and education per child: the *quantity-quality choice*. The model can have two stable regimes: **high fertility and stagnation**, or **lower fertility and sustained growth**. Escaping stagnation requires a sufficiently large shock.

4.4.1. Demographics, state and choices

People live for two periods: childhood, then adulthood. Adults work and consume, and spend time raising and teaching the next generation.

- State: acquired human capital $H \geq 0$. Every adult also has raw-labor productivity $H_0 > 0$; total productivity is $H_0 + H$.

- Choices: consumption $c > 0$, number of children $n > 0$ (continuous), and teaching time per child $h \geq 0$.
- Next state: each child inherits the same human capital H'.
- Adult time endowment is one. Each child requires fixed childcare time $v > 0$ and a goods cost $f \geq 0$.

This version has no assets or aggregate shocks. H_0 is fixed raw-labor productivity, distinct from the initial value of H.

4.4.2. Dynastic preferences

Parents value their own consumption and their children's utility:

$$\begin{aligned} u(c) &= \frac{c^\sigma}{\sigma}, \quad 0 < \sigma < 1, \\ a(n) &= \alpha n^{-\varepsilon}, \quad \alpha > 0, \quad 0 < \varepsilon < 1, \\ V(H) &= u(c) + a(n) n V(H') \\ &= \frac{c^\sigma}{\sigma} + \alpha n^{1-\varepsilon} V(H'). \end{aligned} \tag{4.1}$$

- Altruism **per child**, $a(n)$, falls with fertility; total weight on children, $\alpha n^{1-\varepsilon}$, rises.
- $\varepsilon = -\frac{d \log a(n)}{d \log n}$ measures how altruism per child falls with fertility. Relative risk aversion is $1 - \sigma$.

Assume finite dynastic utility. The continuation weight varies with n.

> **Lecture addition: The continuation value belongs to the child**
>
> $V(H')$ is the child's adult utility, including that child's concern for the next generation. Parents care about their children's actual welfare. If H' differs from parental human capital, the child can choose different fertility and education. Identical people within a generation need not make identical choices across generations.
>
> Recording: 2026-10-12, 01:06:07 - 01:08:09; 01:11:08 - 01:11:59

4.4.3. Resources and human-capital production

Childcare and teaching leave $1 - n(v + h)$ units of working time:

$$\begin{aligned} c + fn &= [1 - n(v + h)](H_0 + H), \\ H' &= Ah(H_0 + H), \quad A > 0, \\ n(v + h) &\leq 1. \end{aligned} \tag{4.2}$$

A is teaching productivity. Higher parental human capital raises both the opportunity cost of time and the amount a parent can teach.

Eliminate h to write the problem in (n, H'):

$$
\begin{aligned}
h &= \frac{H'}{A(H_0 + H)}, \\
c(H, n, H') &= (1 - nv)(H_0 + H) - \frac{nH'}{A} - fn.
\end{aligned}
\tag{4.3}
$$

The Bellman equation is

$$
V(H) = \max_{n>0, H' \geq 0} \left\{ \frac{c(H, n, H')^{\sigma}}{\sigma} + \alpha n^{1-\varepsilon} V(H') \right\}, \quad c(H, n, H') > 0. \tag{4.4}
$$

More children raise the total cost of providing a given education per child.

4.4.4. Optimal education: corner condition and Euler equation

At a differentiable optimum, the marginal benefit cannot exceed the marginal cost at zero investment; they are equal at positive investment:

$$
\begin{aligned}
\alpha n^{1-\varepsilon} V'(H') &\leq \frac{nc^{\sigma-1}}{A}, \\
H' \left[\frac{nc^{\sigma-1}}{A} - \alpha n^{1-\varepsilon} V'(H') \right] &= 0.
\end{aligned}
\tag{4.5}
$$

The envelope condition is

$$
V'(H) = c^{\sigma-1}(1 - nv). \tag{4.6}
$$

Holding H' fixed keeps teaching cost $n\frac{H'}{A}$ independent of H. The envelope coefficient $1 - nv$ therefore differs from working time $1 - n(v + h)$.

Let c' and n' denote the child's adult choices. Substitute the child's envelope condition into the education FOC:

$$
\begin{aligned}
\alpha n^{1-\varepsilon} (c')^{\sigma-1} (1 - n'v) &\leq \frac{nc^{\sigma-1}}{A}, \\
\underbrace{\frac{n^{\varepsilon}}{\alpha} \left(\frac{c'}{c} \right)^{1-\sigma}}_{\text{consumption MRS}} &\geq \underbrace{A(1 - n'v)}_{\text{education return}}.
\end{aligned}
\tag{4.7}
$$

Equality holds if $h > 0$. High fertility raises $\frac{n^{\varepsilon}}{\alpha}$ and discourages investment in each child's future consumption.

Lecture addition: Why education can be zero

Children receive H_0 without teaching. When acquired human capital is low, a small addition to H' gives a small proportional gain in their future productivity, while teaching immediately reduces parental working time. Because parents also pay the fixed food cost fn, the percentage loss in their consumption can exceed the percentage loss in earnings.

When acquired skills account for most of children's future earnings, education has a much larger effect on their welfare. This is the intuition behind the two investment regimes.

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4.4.5. Optimal fertility

For an interior fertility choice, differentiate the Bellman equation with respect to n:

$$\underbrace{(1-\varepsilon)\alpha n^{-\varepsilon}V(H')}_{\text{benefit of another child}} = \underbrace{c^{\sigma-1}[(v+h)(H_0+H)+f]}_{\text{goods and time cost}}. \tag{4.8}$$

Higher H makes childcare and teaching time more expensive. The substitution effect favors fewer children and more education per child. Higher income also raises demand for children, so the net fertility response depends on both effects.

Lecture addition: Positive fertility and the quantity-quality choice

At a fixed continuation value $V(H') > 0$, the marginal benefit of children diverges as fertility approaches zero:

$$(1-\varepsilon)\alpha n^{-\varepsilon}V(H') \to \infty \quad \text{as} \quad n \to 0.$$

This supports positive fertility under the model's assumptions, even when teaching is zero.

Planning more education per child raises the time cost of every additional child through $(v+h)(H_0+H)$. Education choices make children more expensive and can reduce fertility.

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4.4.6. Malthusian steady state

Every generation chooses zero education:

$$\begin{aligned} H = H' = h = 0, \\ c_M = H_0 - n_M(vH_0 + f) > 0. \end{aligned} \tag{4.9}$$

With constant consumption and fertility,

$$\begin{aligned} V(0) &= \frac{c_M^\sigma}{\sigma} + \alpha n_M^{1-\varepsilon} V(0), \\ &= \frac{c_M^\sigma}{\sigma\left[1 - \alpha n_M^{1-\varepsilon}\right]}. \end{aligned} \tag{4.10}$$

Finite utility requires $\alpha n_M^{1-\varepsilon} < 1$. Fertility n_M satisfies the FOC below; substituting $V(0)$ gives its second form:

$$(1-\varepsilon)\alpha n_M^{-\varepsilon} V(0) = c_M^{\sigma-1}(vH_0 + f). \tag{4.11}$$

$$\frac{(1-\varepsilon)\alpha n_M^{-\varepsilon} c_M}{\sigma\left[1 - \alpha n_M^{1-\varepsilon}\right]} = vH_0 + f. \tag{4.12}$$

For a **strict no-investment corner**, the Euler condition becomes

$$\frac{n_M^\varepsilon}{\alpha} > A(1 - n_M v). \tag{4.13}$$

- High fertility raises the left side and lowers the right side. Suitable v, f and preference parameters support the no-investment corner.
- When the zero-investment policy is strictly optimal and remains so near $H = 0$, small increases in H lead back to $H' = 0$ in one generation. The Euler inequality alone is a local condition, not a global optimality proof.
- Consumption per adult stays constant. Under the model's one-adult household convention, $N_{t+1} = n_M N_t$; population grows if $n_M > 1$.

4.4.7. Balanced growth path

For sufficiently large H, education can sustain growth. Along a growing path, $\frac{H_0}{H} \to 0$ and $\frac{f}{c} \to 0$. Set $H_0 = f = 0$ in the limiting problem:

$$\begin{aligned} c &= [1 - n(v+h)]H, \\ H' &= AhH. \end{aligned} \tag{4.14}$$

On the **balanced growth path** (BGP), n and h are constant. Define the gross growth factor $1 + g$:

$$1 + g = \frac{c'}{c} = \frac{H'}{H} = Ah. \tag{4.15}$$

Homogeneity implies $V(H) = CH^\sigma$, with $C > 0$, so

$$\frac{V'(H')H'}{V(H')} = \sigma. \tag{4.16}$$

The interior FOCs for teaching and fertility are

$$\begin{aligned} A\alpha n^{-\varepsilon} V'(H') &= c^{\sigma-1}, \\ (1-\varepsilon)\alpha n^{-\varepsilon} V(H') &= c^{\sigma-1}(v+h)H. \end{aligned} \tag{4.17}$$

Divide the first by the second, then multiply by H':

$$\begin{aligned} \frac{1}{1-\varepsilon}\frac{V'(H')H'}{V(H')} &= \frac{H'}{AH(v+h)}, \\ \frac{\sigma}{1-\varepsilon} &= \frac{h}{v+h}, \\ (1-\varepsilon-\sigma)h &= \sigma v. \end{aligned} \tag{4.18}$$

$$\begin{aligned} h &= \frac{\sigma v}{1-\varepsilon-\sigma}, \\ 1+g &= \frac{A\sigma v}{1-\varepsilon-\sigma}. \end{aligned} \tag{4.19}$$

Positive teaching requires $1-\varepsilon-\sigma > 0$; positive growth requires $1+g > 1$. These conditions describe the limiting growth regime. An economy starting near $H = 0$ can still remain in stagnation.

Fertility on the BGP satisfies the Euler equation

$$\begin{aligned} \frac{n^{\varepsilon}}{\alpha}(1+g)^{1-\sigma} &= A(1-nv), \\ n^{\varepsilon} &= \alpha A(1-nv)(1+g)^{\sigma-1}. \end{aligned} \tag{4.20}$$

The candidate must also satisfy

$$\begin{aligned} n(v+h) &< 1, \\ \alpha n^{1-\varepsilon}(1+g)^{\sigma} &< 1. \end{aligned} \tag{4.21}$$

The first condition leaves time for production and positive consumption. The second ensures finite dynastic utility as consumption grows:

$$V(H) = \frac{[1-n(v+h)]^{\sigma} H^{\sigma}}{\sigma\left[1-\alpha n^{1-\varepsilon}(1+g)^{\sigma}\right]}. \tag{4.22}$$

- On an interior feasible BGP, higher fixed childcare time v raises teaching time per child and the growth rate.
- α affects BGP fertility; it drops out of the growth formula.
- Growth is **endogenous**: parents choose teaching while A stays constant. With $H_0 > 0$ or $f > 0$, the BGP formulas hold asymptotically.

Lecture addition: Mass schooling and fertility

The lecture linked falling fertility in today's high-income countries to the expansion of mass schooling in the late nineteenth century. Education increased the investment families made in each child, motivating the quantity-quality mechanism in these growth models.

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4.4.8. Two regimes and the transition threshold

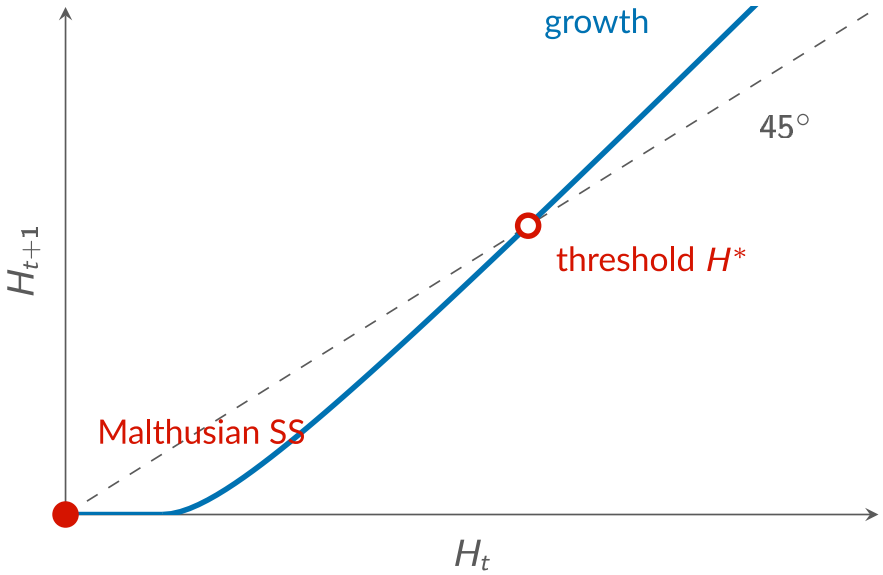


Figure 6: Human capital and the two regimes.

Note: H_t and H_{t+1}: acquired human capital per adult in successive generations (model units). Dashed line: $H_{t+1} = H_t$. H^*: threshold separating stagnation and growth in the schematic. Lecture 5, slide 27. The curve is a schematic, not a calibrated policy function.

- Near zero, $h = 0$ and $H_{t+1} = 0$.
- Below the threshold, positive education can still be insufficient to replace parental human capital: $H_{t+1} < H_t$.
- Above the threshold, $H_{t+1} > H_t$; the economy approaches the BGP. Fertility is lower and teaching per child is higher in the growth regime.
- A large enough shock to human capital or technology can change the eventual regime. A small temporary increase in H may leave it unchanged.

On the BGP, **growth rates and time allocation** stay constant while human capital keeps rising. The smooth threshold crossing is schematic: a nonconcave problem can have a policy jump or a kink in V. FOCs alone do not establish the global optimum.

5. Lecture 5 Oct 13

5.1. Becker, Murphy and Tamura: assessment

- Cheap parental time favors many children. Costly time and productive teaching favor more investment per child.
- The *Malthusian* regime has high fertility and stagnant income per adult. Land is absent, and there is no exogenous productivity trend whose gains are absorbed by population growth.
- A large shock can start sustained growth. Its historical origin is outside the model.
- Stagnation can be optimal for a dynasty under these preferences and constraints. Pareto efficiency across generations requires a specified welfare criterion; it does not follow from household optimality alone.
- The dynastic-altruism framework remains widely used to model fertility in macroeconomics.

5.2. Unified growth theory: Galor and Weil (2000)

Population, Technology, and Growth: From Malthusian Stagnation to the Demographic Transition and Beyond (Galor & Weil, 2000). Lecture 5, slides 29 to 33.

Population growth speeds up technological progress. Faster technological change raises the return to education, shifting parents toward fewer, better-educated children. The transition to modern growth is **endogenous**.

5.2.1. Households and child quality

People live for two periods: childhood, then adulthood. Each adult has one unit of time and chooses consumption c_t, fertility n_t, and education per child $e_{t+1} \geq 0$.

- L_t: adult population; land X is fixed.
- A_t: technology; $g_{t+1} = \frac{A_{t+1}}{A_t} - 1$ is technological progress per generation.
- h_t: human capital per adult; w_t: wage per efficiency unit of labor. Potential income, before childcare, is $z_t = w_t h_t$.
- Each child costs $\tau^q > 0$ units of parental time; each unit of education costs another $\tau^e > 0$ units. Write $\tau(e) = \tau^q + \tau^e e$.

Preferences and the budget constraint are

$$\begin{aligned} u_t &= (1-\gamma)\log c_t + \gamma \log(n_t h_{t+1}), \\ c_t + z_t n_t \tau(e_{t+1}) &= z_t, \\ c_t &\geq \tilde{c} > 0, \quad 0 < \gamma < 1. \end{aligned} \tag{5.1}$$

Parents value the number and human capital of their children. Here γ is the preference weight on children, and $\tilde{c}$ is subsistence consumption. The lecture suppresses the child's future wage in utility: an individual parent takes it as given, so it does not affect the choices.

Lecture addition: Child quality in preferences

Becker, Murphy and Tamura value children's continuation utility. Galor and Weil put $n_t h_{t+1}$ directly into parental utility. This reduced-form specification avoids solving the children's dynastic problem.

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5.2.2. Fertility and subsistence

If the subsistence constraint is slack, the fertility FOC gives

$$\begin{aligned} \frac{\gamma}{n_t} &= \frac{(1-\gamma) z_t \tau(e_{t+1})}{c_t}, \\ n_t \tau(e_{t+1}) &= \gamma, \\ c_t &= (1-\gamma) z_t. \end{aligned} \tag{5.2}$$

This requires $z_t \geq \tilde{z} := \frac{\tilde{c}}{1-\gamma}$. For $z_t > \tilde{c}$, the full solution is

$$n_t \tau(e_{t+1}) = \begin{cases} 1 - \frac{\tilde{c}}{z_t} & \text{if } \tilde{c} < z_t < \tilde{z} \\ \gamma & \text{if } z_t \geq \tilde{z} \end{cases}. \tag{5.3}$$

- Below $\tilde{z}$, $c_t = \tilde{c}$. Higher income frees time from work and raises fertility, holding education fixed.
- Above $\tilde{z}$, the fraction of time spent on children is constant: $n_t = \frac{\gamma}{\tau^q + \tau^e e_{t+1}}$. More income alone does not raise fertility; more education lowers it.

Lecture addition: Population growth requires a parameter restriction

At zero education, unconstrained fertility is

$$n_t^{\text{unconstrained}} = \frac{\gamma}{\tau^q}.$$

Population growth in this regime requires $\frac{\gamma}{\tau^q} > 1$. A ratio below one gives declining population even when subsistence is slack. The income-fertility feedback comes from **binding subsistence**; unconstrained fertility is constant at fixed education.

Recording: 2026-10-13, 00:12:29 - 00:13:49

5.2.3. Education and technological change

Children's human capital is

$$\begin{aligned} h_{t+1} &= h(e_{t+1}, g_{t+1}), \quad h > 0, \\ h_e > 0, \quad h_{ee} < 0, \quad h_g &< 0, \quad h_{eg} > 0. \end{aligned} \tag{5.4}$$

Education raises human capital with diminishing returns. Technological change makes existing skills less useful; education reduces this loss. The paper assumes the productivity gain still raises potential earnings.

After substituting optimal fertility, the education choice in either income regime is

$$e_{t+1} \in \arg\max_{e \geq 0} \frac{h(e, g_{t+1})}{\tau^q + \tau^e e}. \tag{5.5}$$

For positive education, the marginal gain in child quality equals the proportional increase in the time cost per child:

$$\frac{h_e(e, g)}{h(e, g)} = \frac{\tau^e}{\tau^q + \tau^e e}. \tag{5.6}$$

At $e = 0$, the left side is no larger than the right side.

Define $F(e, g) = \tau(e) h_e(e, g) - \tau^e h(e, g)$. At an interior optimum,

$$\begin{aligned} F(e, g) &= 0, \\ F_e &= \tau(e) h_{ee} < 0, \\ F_g &= \tau(e) h_{eg} - \tau^e h_g > 0, \\ \frac{de(g)}{dg} &= -\frac{F_g}{F_e} > 0. \end{aligned} \tag{5.7}$$

The model assumes a threshold $\hat{g}$ satisfying $F(0, \hat{g}) = 0$:

$$\begin{aligned} &e_{t+1} = e(g_{t+1}), \\ &e(g) = 0, \quad \text{if } g \leq \hat{g}, \\ &e(g) > 0, \quad e'(g) > 0, \quad \text{if } g > \hat{g}. \end{aligned} \tag{5.8}$$

Slow technological change gives an **education corner**. Faster change makes education worth its time cost, even though preferences stay fixed.

5.2.4. Technology and population dynamics

Technological progress depends on the education of the current adults and the size of the population:

$$\begin{aligned} g_{t+1} &= g(e_t, L_t), \\ A_{t+1} &= (1 + g_{t+1}) A_t, \\ e_{t+1} &= e(g_{t+1}), \\ h_{t+1} &= h(e_{t+1}, g_{t+1}), \\ L_{t+1} &= n_t L_t. \end{aligned} \tag{5.9}$$

The technology function is increasing in both arguments: educated adults generate more ideas, and a larger population means more potential inventors. $n_t = 1$ means replacement fertility under this population normalization. Each t is a generation, not a year.

Parents take z_t and g_{t+1} as given when choosing n_t and e_{t+1}. Their children's education affects technological progress in $t + 2$. Fixed land holds down income per adult as population expands, unless technology and human capital grow fast enough to offset it.

Lecture addition: The population relevant for scale effects

The relevant population is the network exchanging ideas. Small countries can use knowledge from abroad, so national population need not predict national technological progress. Kremer's historical comparisons concern connected and isolated populations. In a small isolated group, a skill can disappear when its few practitioners die.

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5.2.5. From stagnation to modern growth

Conditional on a fixed L, the education-technology intersections satisfy

$$e^* = e(g^*), \quad g^* = g(e^*, L). \tag{5.10}$$

These are **conditional steady states** for (e, g); technology levels and population need not be constant along the full transition.

1. **Malthusian regime.** Small L gives slow technological progress and $e = 0$, the unique stable conditional steady state. Technology gains mostly raise population; income per adult remains near subsistence. Population growth gradually raises $g(0, L)$.
2. **Post-Malthusian regime.** Larger L accelerates technological progress. The diagram has multiple conditional steady states; the economy initially remains near the low-education one. Income per adult starts rising, but population growth still absorbs part of the output gain. While subsistence binds, higher income also raises the time available for children.
3. **Modern growth regime.** Once $g(0, L) > \hat{g}$, zero education cannot remain optimal. Education and technological progress reinforce each other. Education per child rises, fertility falls, and income per adult grows persistently.

Rising population shifts the technology locus to the right in the lecture's (g, e) diagram. The low-education steady state eventually disappears.

5.2.6. Assessment

- Rising returns to education drive the demographic transition. Above subsistence, income alone leaves fertility unchanged at fixed education.
- The transition can occur without a large outside shock to human capital. It relies on the assumed population scale effect in technological progress.
- Technological progress is a reduced-form function of education and population. Structural change and institutions are outside these equations; the model is difficult to quantify against historical data.

Lecture addition: Structural change and the demand for literacy

On a family farm, practical knowledge can pass directly from parents to children. Large organizations require bookkeeping, accounting and written instructions. Even a delivery driver needs to read addresses and instructions. Changing production organization can therefore raise the return to literacy and formal schooling.

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5.3. Dynamics of education and technology

Hold L fixed within each panel:

- Blue solid curve: parents' education choice, $e = e(g)$.
- Orange dashed curve: technological progress, $g = g(e, L)$.
- Filled points are locally stable; hollow points are unstable.

An intersection is a conditional steady state. Education evolves according to

$$e_{t+1} = e(g(e_t, L)). \tag{5.11}$$

At a positive-education intersection, the response to a small change in e_t is

$$\frac{de_{t+1}}{de_t} = e'(g^*)g_e(e^*, L). \tag{5.12}$$

The steady state is locally stable if this response is below one, unstable if it exceeds one. A larger population shifts the technology locus right; the education-choice curve stays fixed.

5.3.1. Malthusian regime

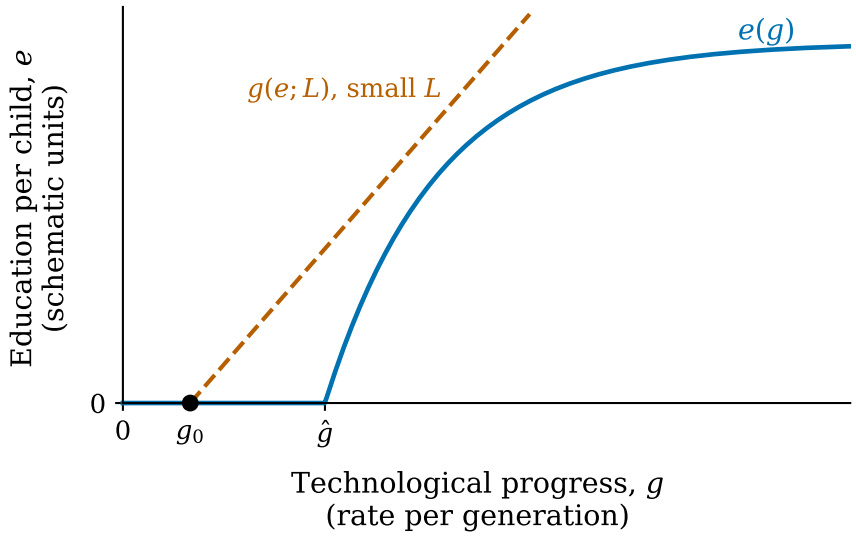


Figure 7: Malthusian regime.

Note: Redrawn schematic based on Lecture 5, slide 32, first overlay. $g_0 = g(0, L) < \hat{g}$; education is zero. The technology locus is adjusted to give the unique intersection described in the slide's text. The curves are illustrative, not calibrated.

- Small population gives slow technological progress. The only conditional steady state is $(g_0, 0)$; small education increases die out.
- Technology can still raise population while education stays zero. As L grows, the technology locus moves right.

5.3.2. Post-Malthusian regime

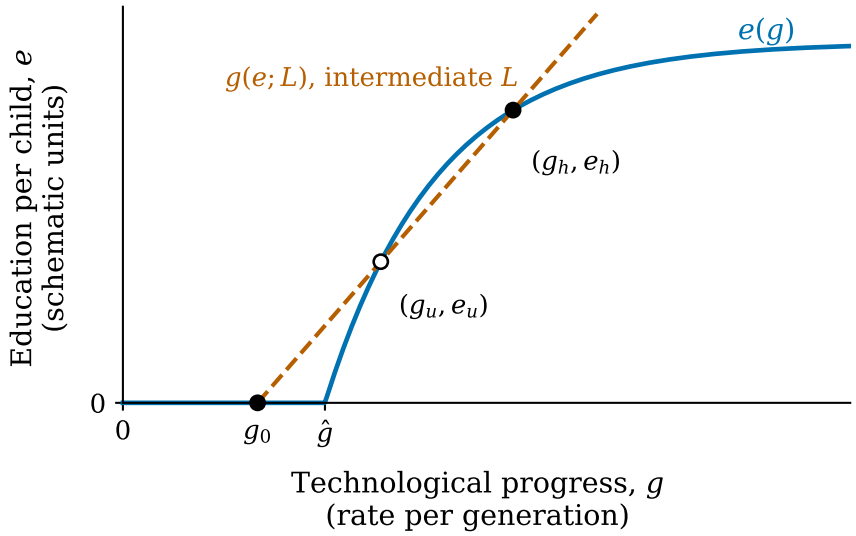


Figure 8: Post-Malthusian regime.

Note: Redrawn schematic based on Lecture 5, slide 32, second overlay. (g_u, e_u) is unstable; (g_h, e_h) is the stable high-education state. The curves are illustrative, not calibrated.

- Three intersections: the zero-education and high-education states are stable; the middle state is unstable.
- At fixed L, e_u separates the two basins of attraction. Below it, education returns to zero; above it, education approaches e_h.
- The economy can remain near the low state while population expands and technology accelerates. The threshold itself moves as L changes.

5.3.3. Modern growth regime

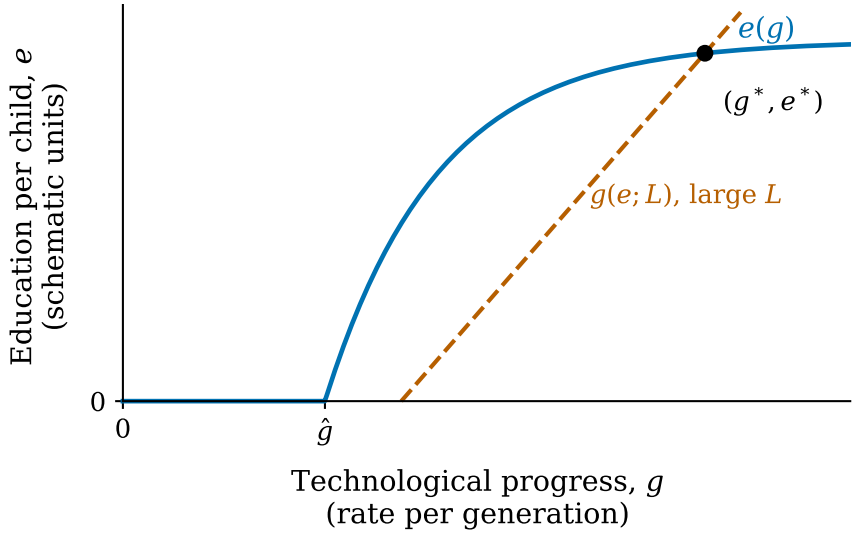


Figure 9: Modern growth regime.

Note: Redrawn schematic based on Lecture 5, slide 32, third overlay. The only intersection is (g^*, e^*), with positive education. The curves are illustrative, not calibrated.

- With $g(0, L) > \hat{g}$, zero education is no longer optimal. The low conditional steady state disappears; education moves toward $e^* > 0$.
- Faster technological change raises education; educated adults generate faster technological progress. Once subsistence is slack, $n_t = \frac{\gamma}{\tau^q + \tau^e e_{t+1}}$: more education lowers fertility.
- The population need not shrink when fertility falls. Income per adult grows as technology and human capital outpace the pressure on fixed land.

5.4. The transition: other contributions

Lecture 5, slide 34.

- (Kremer, 1993): *more people generate more nonrival ideas; better technology supports a larger population under the Malthusian land constraint. The model predicts population growth increasing with population size over much of history. Historical population data and comparisons of technologically isolated societies support this scale effect.*
- (Jones, 2001): *combines idea production, endogenous fertility and mortality, and fixed land. Under suitable returns to accumulable factors, population and ideas can generate a takeoff followed by a demographic transition. In the quantitative model, stronger rewards to inventors, interpreted as improvements in innovation-promoting institutions, substantially advance the takeoff. Its timing depends on these incentives.*
- (Galor & Moav, 2002): *preferences for child quality differ across heritable types. In the Malthusian regime, quality-oriented families can earn more and afford more children, giving their type an evolutionary advantage. Their growing population share raises investment in human capital and technological progress. The distribution of preferences therefore evolves within the model.*
- (Greenwood & Seshadri, 2002): *a quantitative model of the U.S. demographic transition with agriculture and manufacturing. Productivity growth raises wages and the time cost of children. As agriculture contracts, parents increasingly educate children for skilled manufacturing work. Costly education favors fewer children; movement from high-fertility rural households to lower-fertility urban households also reduces aggregate fertility.*
- (Doepke, 2004): *a quantitative transition model linking structural change, fertility, education, and child labor. Restricting child labor removes children's potential earnings, raising the net cost of having many children. In the calibrated model, child labor restrictions have much larger effects on fertility decline than education subsidies. Policies also affect the evolving income distribution.*
- (Galor & Mountford, 2008): *trade changes the returns to child quantity and quality through specialization. Industrial economies specialize in goods intensive in human capital, encouraging education and lower fertility. Agricultural exporters face weaker incentives to educate and can have a delayed fertility transition. Trade can widen income and population-growth differences across countries.*

5.5. The transition: recent research

Lecture 5, slide 35.

5.5.1. Human capital and the Industrial Revolution

- (Squicciarini & Voigtländer, 2015): *distinguishes average literacy from upper-tail knowledge. Subscriptions to the eighteenth-century Encyclopédie proxy the presence of knowledge elites in French cities. Subscriber density predicts growth after industrialization begins. Initial literacy is associated with development levels, but does not predict growth. The mechanism is higher productivity in innovative industrial technology.*
- (de la Croix et al., 2018): *emphasizes tacit knowledge passed from master to apprentice. Institutions determine who can learn from whom. Their model shows how guilds, journeymanship, and apprenticeship markets spread skills beyond closed family or clan networks, helping explain Europe's preindustrial technological advantage.*
- (Kelly et al., 2023): *studies English counties between the 1760s and 1830s. High initial mechanical skills and low wages predict industrialization. Literacy, banks, and proximity to coal have little explanatory power in their regressions.*

5.5.2. Unified growth and cross-country evidence

(Cervellati et al., 2023)

The model combines endogenous physical capital, human capital, population, and technology. Physical and human capital are complementary; technology diffuses internationally. A quantitative multi-country version matches several patterns of long-run growth and comparative development within the same model.

Growth accelerates around the demographic transition and can temporarily exceed its eventual BGP rate as factors accumulate. The model also reproduces patterns in cross-country growth regressions using simulated panel data. Different transition dates help explain differences in income and growth across countries.

5.5.3. Demographic transitions across time and space

(Delventhal et al., 2024)

*The study covers birth and death rates for **186 countries over more than 250 years** (CEMFI Working Paper 2402). The transition from high fertility and mortality to low fertility and mortality is completed or underway in nearly every country.*

- *Later demographic transitions are faster.*
- *Having more neighbors that have started the transition is associated with a higher probability of starting one's own transition.*

Their quantitative model combines parents' quantity-quality choices with the geographic diffusion of production and medical technologies from Great Britain. It matches the differing transition dates, increasing transition speeds, and the association between faster fertility decline and rising schooling.

Lecture addition: What explains recent fertility declines?

- Rising schooling and declining child labor increase the cost of children relative to their earnings contribution.
- Mothers' labor-force participation can raise the time cost of children. The lecture noted rapid fertility decline in several countries without the large participation increase seen during the Western transition.
- Exposure to different family norms through modern media was suggested as a possible additional mechanism. This was an open hypothesis.

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5.6. From Solow to Lucas

Structural change and endogenous population growth can account for the transition from Malthusian stagnation to Solow-style growth. We still need to explain large, persistent differences in income per worker across countries. Lecture 5, slides 36 to 37.

5.6.1. Predictions of the capital-only model

The lecture keeps Cobb-Douglas production as the benchmark:

$$\begin{aligned} Y &= AK^{\alpha}N^{1-\alpha}, \quad 0 < \alpha < 1, \\ y &:= \frac{Y}{N} = A\left(\frac{K}{N}\right)^{\alpha}, \\ \text{MPK} &= \frac{\partial Y}{\partial K} = \alpha A\left(\frac{K}{N}\right)^{\alpha-1}. \end{aligned} \tag{5.13}$$

Here A is **total factor productivity**. The labor-augmenting A_t in the earlier Solow section would enter this expression as $A_t^{1-\alpha}$. Competitive factor income shares are α and $1-\alpha$.

1. **Conditional convergence.** With the same productivity path, saving rate, population growth, and depreciation, countries below the common BGP grow faster through capital accumulation. Different fundamentals can instead imply different BGP targets. Controlling only for population growth does not establish a common target.
2. **Capital returns.** At a given A and labor quality, lower $\frac{K}{N}$ raises MPK because $\alpha - 1 < 0$. Explaining poverty entirely by low physical capital therefore predicts high capital returns in poor countries.

The lecture's puzzles are slow catch-up and persistent income gaps, together with the absence of the very large return differences implied by capital scarcity alone. Allowing different A can fit income differences, but leaves their source unexplained.

Lecture addition: The capital-flow puzzle

If countries differ only in physical capital, high MPK should draw capital toward poor countries. The lecture contrasts this prediction with capital flows concentrated within and between rich countries. Explaining income gaps by physical-capital scarcity alone also has to explain these flows.

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5.6.2. What can explain the income gaps?

- **Human capital:** workers differ in skills and productive knowledge, accumulated through education and learning. Omitting these inputs puts their contribution into measured TFP.
- **Endogenous productivity:** incentives to create and adopt ideas determine technological progress. R&D-based models are the next lecture's topic.
- **Misallocation:** frictions prevent labor and capital from reaching their most productive uses, reducing aggregate output for given resources.

Human capital is an input that can be accumulated (Lucas, 1988). *With productive human capital h per worker,*

$$\begin{aligned} Y &= AK^{\alpha}(hN)^{1-\alpha}, \\ \text{MPK} &= \alpha A\left(\frac{K}{hN}\right)^{\alpha-1}. \end{aligned} \tag{5.14}$$

The relevant scarcity is now $\frac{K}{hN}$. At a common A, increasing $\frac{K}{N}$ and h by the same proportion raises income per worker by that proportion while leaving MPK unchanged.

5.7. Main idea of Lucas (1988)

On the Mechanics of Economic Development (Lucas, 1988). Lecture 5, slides 38 to 40 use a simplified discrete-time planner problem without human-capital externalities.

Human capital is a factor that can be accumulated. The worker divides productive capacity between making goods and learning. Both technology parameters, A and B, are constant.

5.7.1. States, choices, and resources

- States: physical capital $k > 0$ and human capital $h > 0$. Population is fixed and normalized to one.
- Choices: consumption $c > 0$, next-period physical capital $k' \geq 0$, and human-capital allocations $h_1, h_2 \geq 0$.
- h_1 is used in goods production; h_2 is used in learning. They satisfy $h_1 + h_2 = h$.
- Next-period human capital is $h' = Bh_2$, where $B > 0$ is learning productivity. This is the whole human-capital transition in this specification.

Preferences are CRRA:

$$u(c) = \begin{cases} \frac{c^{1-\sigma}}{1-\sigma} & \text{if } \sigma \neq 1 \\ \log c & \text{if } \sigma = 1 \end{cases}, \quad \sigma > 0. \tag{5.15}$$

Assume $0 < \beta < 1$, $0 < \alpha < 1$, $A > 0$, and $0 \leq \delta \leq 1$. Here σ is relative risk aversion.

5.7.2. Bellman equation

The social planner solves

$$v(k, h) = \max_{c,k',h_1,h_2} \{u(c) + \beta v(k', Bh_2)\} \tag{5.16}$$

subject to

$$\begin{aligned} c + k' &\leq (1 - \delta)k + Ak^{\alpha} h_1^{1-\alpha}, \\ h' &= Bh_2, \\ h &= h_1 + h_2, \\ c &> 0, \quad k', h_1, h_2 \geq 0. \end{aligned} \tag{5.17}$$

The goods constraint binds because utility is increasing in consumption. More h_1 raises current output; more h_2 raises next period's productive capacity.

Lecture addition: Why exact linearity matters

Learning mainly uses learners' and teachers' time. In the EC442 example, forgone earnings are the main cost; classroom equipment is secondary.

To see the role of **linearity**, hold the learning share $s = \frac{h_2}{h}$ fixed and compare a diminishing-returns technology:

$$\begin{aligned} h' &= Bh_2^{\eta}, \quad 0 < \eta < 1, \\ \frac{h'}{h} &= Bs^{\eta} h^{\eta - 1}. \end{aligned}$$

This growth factor falls as h rises. The linear case $\eta = 1$ preserves a constant return and permits sustained growth from human-capital accumulation.

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5.7.3. Optimality conditions

Write goods production and its marginal products as

$$\begin{aligned} f(k, h_1) &= Ak^{\alpha} h_1^{1-\alpha}, \\ f_k(k, h_1) &= \alpha A \left(\frac{h_1}{k} \right)^{1-\alpha}, \\ f_{h_1}(k, h_1) &= (1 - \alpha) A \left(\frac{k}{h_1} \right)^{\alpha}. \end{aligned} \tag{5.18}$$

Eliminating h_2 gives $h_1 = h - \frac{h'}{B}$. At an interior optimum, the FOCs for k' and h' are

$$\begin{aligned} u'(c) &= \beta v_k(k', h'), \\ u'(c) f_{h_1}(k, h_1) &= \beta B v_h(k', h'). \end{aligned} \tag{5.19}$$

The second condition equates the output lost by shifting human capital into learning with the discounted value of the human capital it creates.

The envelope conditions are

$$\begin{aligned} v_k(k,h) &= u'(c)[1-\delta+f_k(k,h_1)], \\ v_h(k,h) &= u'(c)f_{h_1}(k,h_1). \end{aligned} \tag{5.20}$$

Substitute next period's envelope conditions into the FOCs and use $u'(c) = c^{-\sigma}$:

$$\begin{aligned} \left(\frac{c'}{c}\right)^{\sigma} &= \beta\left[1-\delta+\alpha A\left(\frac{h_1'}{k'}\right)^{1-\alpha}\right], \\ \left(\frac{c'}{c}\right)^{\sigma} &= \beta B\left(\frac{k'}{h_1'}\right)^{\alpha}\left(\frac{h_1}{k}\right)^{\alpha}. \end{aligned} \tag{5.21}$$

Here c' and h_1' are next period's consumption and production allocation. The first equation governs physical investment; the second governs learning. The learning return includes the change in human capital's marginal product.

5.7.4. Balanced growth path

On a BGP, allocation shares and capital ratios are constant:

$$\gamma = \frac{c'}{c} = \frac{k'}{k} = \frac{h'}{h} = \frac{h_1'}{h_1}. \tag{5.22}$$

Here γ is the **gross growth factor**; net growth is $\gamma - 1$. In the learning Euler equation, the two capital-ratio terms cancel:

$$\begin{aligned} \gamma^{\sigma} &= \beta B \gamma^{\alpha}\gamma^{-\alpha} = \beta B, \\ \gamma &= (\beta B)^{\frac{1}{\sigma}}. \end{aligned} \tag{5.23}$$

Sustained positive growth requires $\beta B > 1$.

Let $r^K = f_k(k, h_1)$ be the capital rental rate, written as r on slide 40. The physical-capital Euler equation implies

$$\begin{aligned} \gamma^{\sigma} &= \beta\left(1-\delta+r^K\right), \\ r^K &= B-1+\delta, \\ 1-\delta+r^K &= B. \end{aligned} \tag{5.24}$$

The gross return to holding physical capital equals B; its net return is $r^K - \delta = B - 1$. The rental rate itself includes depreciation compensation.

Human-capital accumulation fixes the allocation shares:

$$\begin{aligned}\frac{h_2}{h} &= \frac{\gamma}{B},\\ \frac{h_1}{h} &= 1 - \frac{\gamma}{B}.\end{aligned} \tag{5.25}$$

An interior growing BGP requires $\gamma < B$, leaving human capital for goods production. For $\sigma \neq 1$, this is also the finite-utility condition along the BGP:

$$\beta\gamma^{1-\sigma} = \frac{\gamma}{B} < 1. \tag{5.26}$$

With log utility, $\gamma = \beta B$ and $\frac{h_2}{h} = \beta$.

The constant physical-to-productive-human-capital ratio satisfies

$$\begin{aligned}\frac{k}{h_1} &= \left(\frac{\alpha A}{B - 1 + \delta}\right)^{\frac{1}{1-\alpha}},\\ \frac{k}{h} &= \left(1 - \frac{\gamma}{B}\right)\left(\frac{\alpha A}{B - 1 + \delta}\right)^{\frac{1}{1-\alpha}}.\end{aligned} \tag{5.27}$$

These ratios require $B - 1 + \delta > 0$, which holds on a growing BGP.

5.7.5. Growth and income differences

- Growth is **endogenous**: k and h accumulate while A and B stay fixed. The growth factor depends on learning productivity, patience, and preferences.
- Along the BGP, higher A changes capital ratios and income levels, but drops out of $\gamma = (\beta B)^{\frac{1}{\sigma}}$.
- With common parameters, countries can have different income levels on parallel BGPs. Proportional differences in k and h leave $\frac{k}{h}$, γ, and r^K unchanged.
- For a given human-capital path, a low initial $\frac{k}{h}$ creates transitional capital deepening, as in Solow. Persistent human-capital gaps can instead sustain different income levels.

> **Lecture addition: Postwar catch-up**
>
> Germany and Japan in 1946 illustrate physical-capital destruction with much of the workforce's knowledge intact. A low $\frac{k}{h}$ then predicts rebuilding and catch-up, conditional on the human-capital path. Countries with persistent skill gaps can instead remain on parallel BGPs.
>
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5.8. How much does human capital explain? Development accounting

Development accounting decomposes cross-country differences in real GDP per worker into inputs and productivity. Lecture 5, slide 42.

5.8.1. Output per worker

For country i, let N_i be the number of workers and h_i human capital per worker. Write Hicks-neutral TFP as T_i, previously denoted by A, and use a common capital share $0 < \alpha < 1$:

$$
\begin{aligned}
Y_i &= T_i K_i^{\alpha} (h_i N_i)^{1-\alpha}, \\
y_i &:= \frac{Y_i}{N_i} = T_i \left(\frac{K_i}{N_i}\right)^{\alpha} h_i^{1-\alpha}.
\end{aligned}
\tag{5.28}
$$

Substitute $\frac{K_i}{N_i} = \left(\frac{K_i}{Y_i}\right) y_i$ and solve for y_i:

$$
\begin{aligned}
y_i^{1-\alpha} &= T_i \left(\frac{K_i}{Y_i}\right)^{\alpha} h_i^{1-\alpha}, \\
y_i &= \underbrace{\left(\frac{K_i}{Y_i}\right)^{\frac{\alpha}{1-\alpha}}}_{\text{capital intensity}} \quad \underbrace{h_i}_{\text{human capital}} \quad \underbrace{T_i^{\frac{1}{1-\alpha}}}_{\text{productivity}}.
\end{aligned}
\tag{5.29}
$$

The slide calls $T_i^{\frac{1}{1-\alpha}}$ simply A_i. Define

$$
x_{K,i} := \left(\frac{K_i}{Y_i}\right)^{\frac{\alpha}{1-\alpha}}, \quad A_i := T_i^{\frac{1}{1-\alpha}}. \tag{5.30}
$$

Then $y_i = x_{K,i} h_i A_i$.

- Measure y_i and $\frac{K_i}{Y_i}$; construct h_i from worker characteristics.
- Back out productivity as the **residual**: $A_i = \frac{y_i}{x_{K,i} h_i}$.
- At fixed $\frac{K_i}{Y_i}$ and h_i, higher T_i raises output and capital per worker. This decomposition assigns both effects to productivity.

5.8.2. Each factor's contribution

(Klenow & Rodríguez-Clare, 1997): *assign each factor its covariance with log income, relative to the total variance of log income.*

Taking logs gives

$$
\log y_i = \log x_{K,i} + \log h_i + \log A_i. \tag{5.31}
$$

For $x_i \in \{x_{K,i}, h_i, A_i\}$, define

$$
\text{share}_x := \frac{\operatorname{cov}(\log x_i, \log y_i)}{\operatorname{var}(\log y_i)}. \tag{5.32}
$$

Covary the log identity with $\log y_i$:

$$
\begin{aligned}
\operatorname{var}(\log y_i) &= \operatorname{cov}\big(\log x_{K,i}, \log y_i\big) + \operatorname{cov}(\log h_i, \log y_i) \\
&\quad + \operatorname{cov}(\log A_i, \log y_i),
\end{aligned}
\tag{5.33}
$$

$$
\text{share}_K + \text{share}_h + \text{share}_A = 1.
$$

The capital contribution uses $\log x_{K,i} = \frac{\alpha}{1-\alpha} \log\left(\frac{K_i}{Y_i}\right)$, so it includes the production elasticity. These are **accounting shares**; they can be negative or exceed one and do not identify causal effects.

5.8.3. Classic human-capital measure

The lecture uses a schooling-based measure:

$$h_i = \exp(0.1S_i), \quad \log h_i = 0.1S_i, \tag{5.34}$$

where S_i is average years of schooling. One extra year raises measured human capital by $\exp(0.1) - 1 \approx 10.5\%$.

- The coefficient 0.1 comes from the log-wage return to schooling. Interpreting wages as proportional to human capital links this return to h_i.
- The same return is used across countries. School quality, experience, and health are omitted from this measure.

> **Lecture addition: From wage returns to productive human capital**
>
> The calculation assumes that schooling-related pay differences reflect productivity. A wage return from signaling need not raise aggregate output. Individual wage returns also need not remain unchanged when the supply of educated workers increases.
>
> Recording: 2026-10-13, 00:46:29 - 00:46:52

(Caselli, 2005): *classic development accounting leaves large productivity differences after measuring physical capital and schooling-based human capital. The residual also absorbs omitted input quality and measurement error.*

5.9. Development accounting revisited: Lagakos and Schoellman (2026)

Lecture 5, slides 43 to 44.

(Lagakos & Schoellman, 2026): *better measurement raises the contribution of inputs from about 30% under the classic specification to 55–70%, mainly through human capital. The paper is forthcoming in the JEL.*

5.9.1. The classic residual

With schooling-only $h_i = \exp(0.1S_i)$, the capital-output specification still assigns about **70%** of cross-country income variation to A_i. The residual is much lower in poor countries than in the U.S.

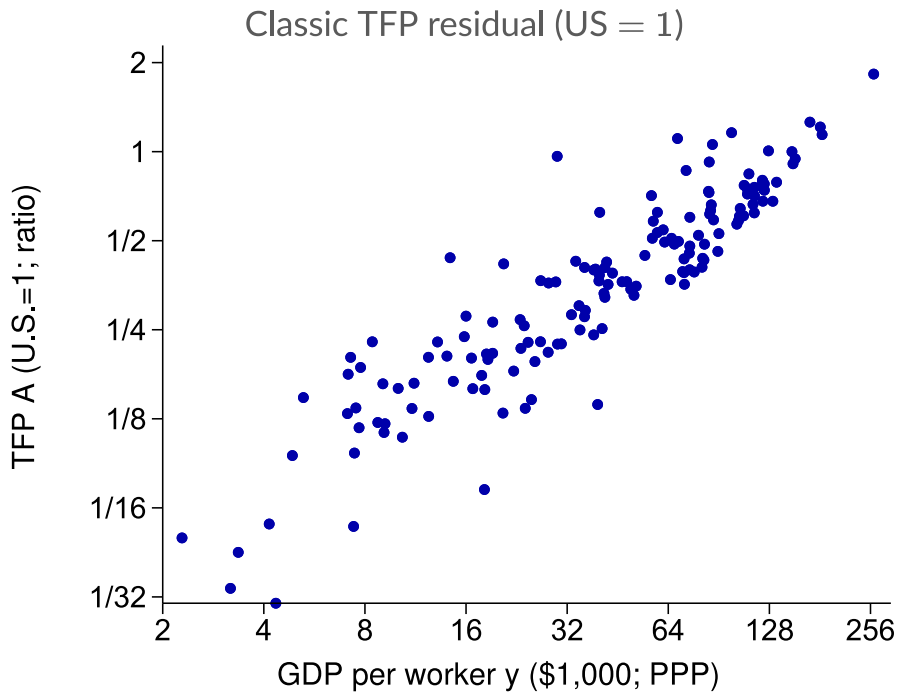


Figure 10: Productivity in the classic specification.

Note: $\frac{A_i}{A_{US}}$: productivity residual, U.S. normalized to one; y_i: real GDP per worker in thousands of PPP dollars. Both axes are logarithmic. Vector crop from Lecture 5, slide 43, based on (Lagakos & Schoellman, 2026); axis notation added.

Allow hours worked per worker, ℓ_i, to enter separately:

$$y_i = x_{K,i} h_i \ell_i A_i, \quad x_{K,i} = \left(\frac{K_i}{Y_i}\right)^{\frac{\alpha}{1-\alpha}}. \tag{5.35}$$

The covariance contributions then satisfy

$$\begin{aligned} \text{share}_{\text{inputs}} &= \text{share}_K + \text{share}_h + \text{share}_\ell, \\ \text{share}_A &= 1 - \text{share}_{\text{inputs}}. \end{aligned} \tag{5.36}$$

- **Hours:** about −5%. Poor-country workers work longer hours, so hours offset part of the income gap; they do not account for low output per worker.
- **Physical capital:** about 7% in this specification. Better measurement changes this contribution little.
- **Human capital:** about 58– 66% across the approaches discussed. The preferred overall input range is 55– 70% across specifications and assumptions.

5.9.2. More dimensions of human capital

Schooling quantity alone misses school quality, experience and health. The lecture's quality adjustment is

$$h_i^{\text{school}} = \exp(0.1 S_i + 0.2 Q_i), \tag{5.37}$$

where Q_i is a standardized test score, in standard-deviation units. At fixed S_i, one extra standard deviation raises this index by $\exp(0.2) - 1 \approx 22.1\%$.

Human-capital measure	Reported contribution
Years of schooling	23%
Schooling plus test-score quality	37%
Experience alone	15%
Health, proxied by height	10%
Joint human-capital measure	66%

These are the rounded figures in slide 44. The quality row already includes schooling. Component measures have different country coverage; the joint estimate is not the sum of these displayed rows.

- Experience human capital reflects both years worked and what workers learn on the job. Wage-experience profiles are steeper in richer countries.
- Height proxies childhood nutrition and health, which affect adult productivity.
- Test scores capture learning from schools and other environments; they are not a measure of school inputs alone.

Lecture addition: Experience and demography

Lower fertility and slower population growth produce an older workforce with more years of experience. Cross-country experience gaps therefore reflect demographic composition as well as differences in learning on the job.

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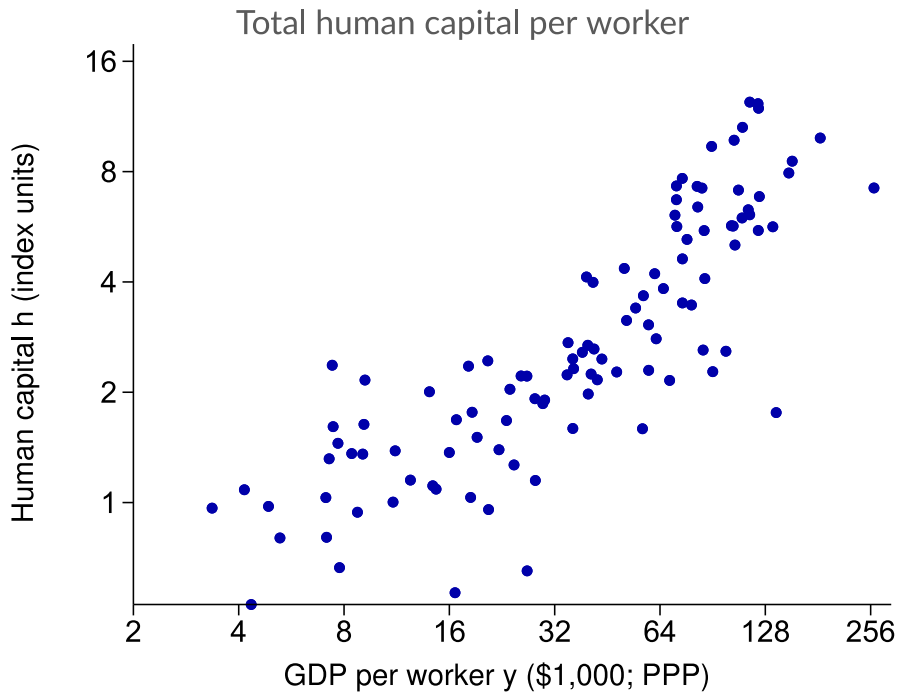


Figure 11: Human capital with broader measurement.

Note: h_i: constructed human-capital index per worker (efficiency units); y_i: real GDP per worker in thousands of PPP dollars. Both axes are logarithmic. Vector crop from Lecture 5, slide 44, based on (Lagakos & Schoellman, 2026); axis notation added.

(Manuelli & Seshadri, 2014): *a structural human-capital model uses schooling and life-cycle earnings profiles to infer human-capital quantities and quality. It finds larger human-capital gaps and smaller TFP gaps than classic measures.*

For given y_i and $x_{K,i}$, increasing measured $h_i \ell_i$ lowers the inferred $A_i = \frac{y_i}{x_{K,i} h_i \ell_i}$. This is a change in measurement and attribution; it does not mean a country's actual technology has changed.

5.10. Evidence from migrants

Lecture 5, slide 45.

(Hendricks & Schoellman, 2018): *compare real, PPP-adjusted hourly wages of the same immigrants before and after moving to the U.S. The main data source is the New Immigrant Survey of new U.S. permanent residents.*

5.10.1. What changes when a worker moves?

The migrant carries **human capital** h_m. The destination supplies its own capital intensity and productivity. In the benchmark, labor markets are competitive, workers are perfect substitutes, the capital share is common, and working hours and the migrant's human capital are held fixed.

Let q_i be the wage per unit of human capital in country i. From $y_i = x_{K,i} h_i A_i$,

$$\begin{aligned} q_i &= (1-\alpha) x_{K,i} A_i, \\ w_{m,i} &= q_i h_m. \end{aligned} \tag{5.38}$$

The same person's human capital cancels from the wage ratio:

$$G_i := \frac{w_{m,\mathrm{US}}}{w_{m,i}} = \frac{q_{\mathrm{US}}}{q_i}. \tag{5.39}$$

Thus **wage gains identify the country component**, combining capital intensity and productivity, under these assumptions.

5.10.2. From wage gains to the human-capital contribution

Define the income and human-capital gaps as

$$D_i := \frac{y_{\mathrm{US}}}{y_i}, \quad H_i := \frac{h_{\mathrm{US}}}{h_i}. \tag{5.40}$$

With a common α,

$$\begin{aligned} D_i &= G_i H_i, \\ \log D_i &= \log G_i + \log H_i, \\ \mathrm{share}_h^{\mathrm{migration}} &= \frac{\log H_i}{\log D_i} = 1 - \frac{\log G_i}{\log D_i}. \end{aligned} \tag{5.41}$$

This is a **log-gap decomposition** relative to the U.S.; the previous section's covariance shares summarize variation across the whole country sample.

For the poorest origin-country group in the lecture, $G_i = 3.2$ and $D_i = 31.8$:

$$H_i = \frac{31.8}{3.2} \approx 9.94,$$
$$\text{share}_h^{\text{migration}} = 1 - \frac{\log 3.2}{\log 31.8} \approx 66\%. \tag{5.42}$$

A threefold wage gain is large for the migrant, but small relative to the thirtyfold income gap. About two-thirds of this group's **log income gap** is attributed to human capital in the benchmark.

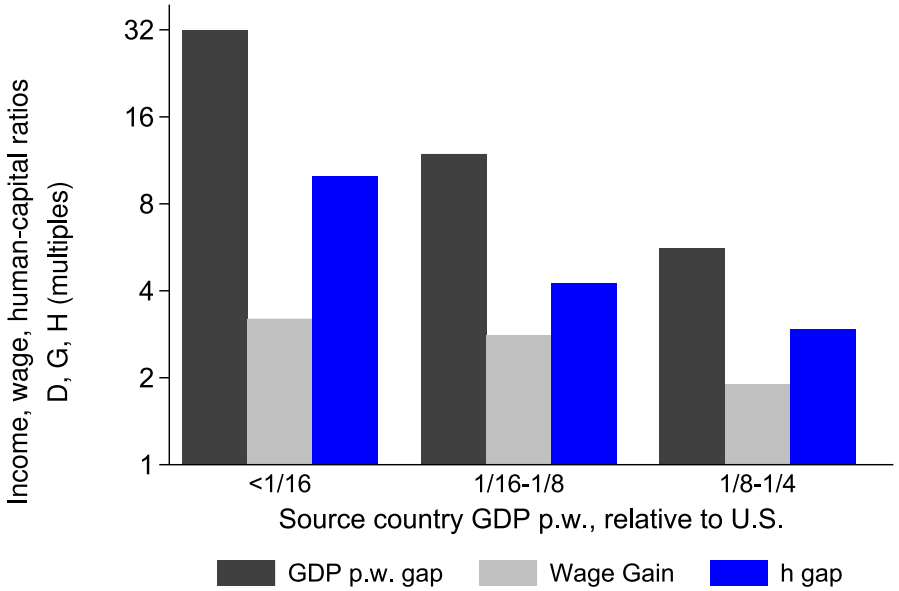


Figure 12: Migrant wage gains and income gaps.

Note: Origin-country bins use $\frac{y_i}{y_{US}}$. Bars show the income gap D_i, wage gain G_i, and inferred human-capital gap $H_i = \frac{D_i}{G_i}$; all are U.S.-to-origin multiples on a logarithmic vertical scale. Cropped from Lecture 5, slide 45. Source: (Lagakos & Schoellman, 2026) (JEL, forthcoming), based on (Hendricks & Schoellman, 2018).

5.10.3. Selection and skill transfer

- **Selection:** migrants need not represent their origin country's average worker. Comparing the same person before and after migration cancels h_m even when migrants are highly selected. Comparing only migrants' U.S. wages across origins does not remove this selection.
- **Skill transfer:** language barriers, unrecognized credentials, or occupational downgrading can reduce effective h_m after migration. Then G_i understates the country component and the formula overstates human capital's contribution.
- **Different skill returns:** wage gains vary with education. Allowing imperfect substitution across skill types gives a human-capital contribution between one-half and two-thirds in the paper, with a benchmark estimate around 60%. The lecture's 66% is the poorest-group calculation, not the full-sample estimate.

5.11. Skilled versus unskilled labor

Lecture 5, slide 46.

(Rossi, 2022): *skilled workers are relatively scarce in poor countries, but their wage premium is not correspondingly higher. Matching quantities and wages requires large differences in relative skill efficiency; migrant evidence attributes most of these differences to skill-biased technology.*

5.11.1. CES labor input and the skill premium

For skill group $j \in \{u, s\}$, let $N_{j,i}$ be workers and $h_{j,i}$ human capital per worker. Effective labor is $L_{j,i} = N_{j,i}h_{j,i}$. The labor aggregate is

$$L_i = \left(z_{u,i} L_{u,i}^{\frac{\sigma-1}{\sigma}} + z_{s,i} L_{s,i}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}. \tag{5.43}$$

Here $z_{j,i} > 0$ are technology weights; σ is the **elasticity of substitution**, distinct from the CRRA parameter used earlier. The displayed formula uses $\sigma > 1$; the lecture's benchmark is $\sigma = 1.5$. A Cobb-Douglas limit at $\sigma = 1$ requires normalized CES weights.

With competitive factor pricing, the common marginal product of aggregate labor cancels from the wage ratio per efficiency unit:

$$\frac{w_{s,i}^e}{w_{u,i}^e} = \frac{z_{s,i}}{z_{u,i}} \left(\frac{L_{s,i}}{L_{u,i}} \right)^{-\frac{1}{\sigma}}. \tag{5.44}$$

Observed pay per worker includes each group's human capital, $W_{j,i} = w_{j,i}^e h_{j,i}$. Thus the **skill premium** is

$$\omega_i := \frac{W_{s,i}}{W_{u,i}} = \frac{z_{s,i}}{z_{u,i}} \left(\frac{h_{s,i}}{h_{u,i}} \right)^{\frac{\sigma-1}{\sigma}} \left(\frac{N_{s,i}}{N_{u,i}} \right)^{-\frac{1}{\sigma}}. \tag{5.45}$$

Define **relative skill efficiency**

$$R_i := \frac{z_{s,i}}{z_{u,i}} \left(\frac{h_{s,i}}{h_{u,i}} \right)^{\frac{\sigma-1}{\sigma}}. \tag{5.46}$$

Solving the wage-premium equation gives

$$R_i = \omega_i \left(\frac{N_{s,i}}{N_{u,i}} \right)^{\frac{1}{\sigma}}. \tag{5.47}$$

If ω_i varies little, greater relative skill supply in rich countries requires higher R_i. The lecture's calculation at $\sigma = 1.5$ implies roughly a sixteenfold rich-poor difference.

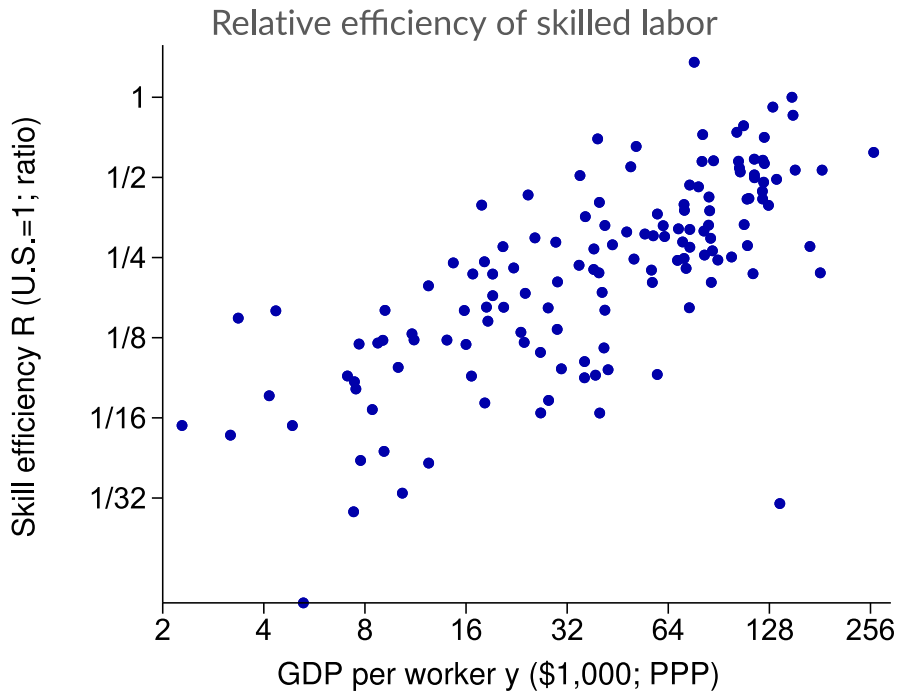


Figure 13: Relative efficiency of skilled labor.

Note: $\frac{R_i}{R_{US}}$: relative skill efficiency, U.S. normalized to one; y_i: real GDP per worker in thousands of PPP dollars. Both axes are logarithmic. Calculations use $\sigma = 1.5$. Vector crop from Lecture 5, slide 46; source: (Lagakos & Schoellman, 2026) and (Rossi, 2022).

5.11.2. Technology versus human capital

The observed premium and worker counts identify R_i, but cannot separately identify $\frac{z_{s,i}}{z_{u,i}}$ and $\frac{h_{s,i}}{h_{u,i}}$. Comparing foreign-educated workers in one destination holds destination technology fixed. Their returns to schooling provide additional information about human-capital quality, subject to selection and skill-transfer assumptions.

The CES labor aggregate combines skills and technology. Its accounting contribution therefore includes skill bias. In the review's illustrative calculation, the contribution of $\frac{L_i}{N_i}$ depends strongly on σ:

Elasticity σ	Accounting contribution of $\frac{L_i}{N_i}$
1.2	100.1%
1.5	51.7%
2.0	35.6%

(Lagakos & Schoellman, 2026), Table 7: *the calculation normalizes unskilled technology and human capital to one and omits experience and health. The share above 100% is permitted by the covariance metric.*

5.12. Human capital and the allocation of talent

Lecture 5, slide 47.

(Hsieh et al., 2019): *in 1960, 94% of U.S. doctors and lawyers were white men; by 2010, 62% were. A Roy model links changing occupational choices to declining barriers and estimates that better allocation accounts for 20–40% of growth in aggregate market output per person over 1960–2010.*

Individuals differ in their comparative advantage across occupations. Their choices also depend on group-specific forces:

- Labor-market discrimination lowers pay relative to marginal product.
- Barriers to acquiring human capital raise the cost of training for an occupation.
- Preferences and social norms affect the utility of market and home occupations.

The model assumes that relative distributions of innate talent across groups are stable over time. Human-capital barriers act through cohort investments; labor-market discrimination affects workers of different cohorts at a given date. Life-cycle wages and occupational shares help distinguish these forces.

Lower barriers let more people enter occupations where their talents are productive and change incentives to acquire skills. Misallocation can therefore lower output even with substantial measured human capital, appearing in the TFP residual. The reported gains concern **market output**; welfare also depends on occupational preferences and home production.

> **Lecture addition: Firm-size barriers and misallocation**
>
> Expanding a business may trigger taxes or compliance costs that can be avoided by staying small. Productive firms then remain undersized, while inputs stay in firms with lower marginal returns. Output falls for given aggregate capital and skills, appearing as lower measured productivity.
>
> Recording: 2026-10-13, 00:57:57 - 00:59:13

5.13. Human capital and ideas

Lecture 5, slide 49.

The lecture's simplified Lucas model has linear private human-capital accumulation. Learning from other people adds an aggregate object: the **distribution of knowledge**. Human capital is embodied in workers; an idea can be used by many workers simultaneously.

- (Lucas & Moll, 2014): *people divide time between producing with their current knowledge and meeting others to learn better ideas. The individual state is productivity z; the aggregate state is its distribution F_t. Search choices change knowledge diffusion and generate endogenous growth. A planner also accounts for the external benefits of learning.*
- (Lucas, 2015): *learning is faster when people interact with more productive people. The model connects individual earnings profiles to aggregate growth.*
- (de la Croix et al., 2018): *apprenticeship institutions determine who learns from whom. Knowledge spreads farther when training extends beyond family networks.*

Lecture addition: Learning from different people

Relatives trained by the same masters may hold similar knowledge. Journeymanship exposes apprentices to different masters and ideas, widening the opportunities to learn.

The modern example is a bank adopting AI by hiring workers who already use it. Knowledge moves between firms with people. The lecture described LinkedIn career histories as data for relating worker flows to technology adoption; no causal identification strategy was discussed.

Recording: 2026-10-13, 01:08:49 - 01:12:23

5.13.1. From learning to innovation

- (Romer, 1990): *ideas are nonrival and partly excludable. Profit incentives finance research; monopoly rights can reward innovation. The resulting increasing returns require a departure from pure price taking.*
- (Jones, 2022): *in idea-based models, a persistently declining population can eventually stop growth in living standards. The number of potential inventors and the demographic response therefore matter for long-run growth.*

5.14. Summary and outlook

Lecture 5, slides 48 and 50.

- The historical transition combines movement away from fixed-land production, a demographic transition and faster productivity growth. Human-capital investment links fertility choices to technological progress.
- Modern measurement and migrant evidence assign roughly three-fifths of cross-country income differences to human capital in the exercises discussed. Schooling-only measures in (Hall & Jones, 1999) assigned about one-fifth. These figures depend on the sample, production function and accounting metric.
- Development accounting locates gaps in inputs and productivity. It does not identify why countries accumulated different skills or which policy would close the gaps. Skill-biased technology also complicates the boundary between human capital and TFP.
- Growth models give mechanisms for accumulation, learning and the transition. Institutions shape investment and innovation incentives. The next lecture turns to the creation of ideas and endogenous productivity growth.

6. Lecture 6 Oct 19

6.1. Endogenous productivity

Physical and human capital explain part of cross-country income differences. We also need to explain **productivity differences** across countries and **productivity growth** over time.

Sources of productivity gains:
- R&D creates new goods or improves existing ones.
- Technology diffusion spreads ideas and production methods across firms and countries.
- Better allocation moves labor and capital toward more productive uses, raising output for given aggregate inputs.

This lecture focuses on **R&D-based growth**:
- (Romer, 1990): *growth through an expanding variety of intermediate goods.*
- (Aghion & Howitt, 1992): *growth through quality improvements. New innovations displace older technologies and their producers: creative destruction.*

> **Lecture addition: Institutions and accumulation**
>
> Explaining why human-capital stocks differ requires a further causal theory. The lecturer pointed to institutions, while emphasizing that this literature has no single canonical model comparable to the growth models studied here.
>
> Recording: 2026-10-19, 00:02:13 - 00:04:19

6.2. (Romer, 1990): Expanding varieties

6.2.1. Main idea

Ideas are **nonrival**: using a blueprint does not prevent others from using it at the same time. Producing the physical good still uses resources.

- Inventing a blueprint requires a **fixed R&D cost**. Once the idea exists, it can be reused without repeating that cost.
- With unrestricted imitation and competitive pricing, innovators may earn too little profit to recover the R&D cost. Useful inventions need not be privately profitable.
- Patents make ideas **partly excludable**: the patent holder can restrict others' use of the invention. In the model, each patent gives a monopoly over one intermediate-good variety.
- Monopoly profits reward innovation. Firms invest in R&D when the expected value of these profits covers the cost of inventing a new variety.

R&D expands the range of intermediate goods. More varieties raise output for given aggregate inputs, linking productivity growth to firms' innovation incentives.

Lecture addition: Access to ideas and research funding

Public funding of universities and research grants can finance invention directly. Imitation delays can also create temporary first-mover rents: copying an idea and building production capacity take time.

A patent can disclose the blueprint publicly while giving its owner exclusive commercial rights. **Nonrivalry** and **excludability** describe different properties of an idea.

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6.2.2. Environment

Time is discrete. The final good is the numeraire and can be consumed, used to produce intermediates, or spent on R&D.

A competitive final-goods sector combines labor and a continuum of intermediate varieties:

$$Y_t = AL^{1-\alpha} \int_0^{N_t} X_{it}^{\alpha} \, \mathrm{d}i, \quad A > 0, \quad 0 < \alpha < 1. \tag{6.1}$$

- $L > 0$: constant labor supply.
- N_t: measure of varieties available at the start of period t.
- $X_{it} \geq 0$: quantity of intermediate i used in production.
- A: constant technology parameter. Innovation increases N_t.

One unit of the final good produces one unit of any existing intermediate, so **marginal cost is one**. Intermediates are used up within the period. Each variety is supplied by a monopolist holding a *permanent patent.*

Inventing one new variety costs $\eta > 0$ units of the final good. With R&D spending $R_t \geq 0$,

$$N_{t+1} = N_t + \frac{R_t}{\eta}. \tag{6.2}$$

Timing: varieties invented in t become available for production in $t + 1$. Existing varieties remain available; blueprints accumulate.

The aggregate resource constraint is

$$Y_t = \underbrace{C_t}_{\text{consumption}} + \underbrace{\int_0^{N_t} X_{it} \, \mathrm{d}i}_{\text{intermediate production}} + \underbrace{R_t}_{\text{R\&D}}, \tag{6.3}$$

where $C_t \geq 0$ is aggregate consumption. R&D uses goods today to expand tomorrow's production possibilities.

Lecture addition: Varieties and production timing

- Differentiated final goods can generate gains from variety in consumption, providing another application of the variety mechanism.
- An integer number of varieties would make changes in N_t discrete. The continuum permits smooth variety expansion.
- Final goods and intermediates are produced simultaneously within the period. Producing intermediates today for use tomorrow would add inventories as state variables.

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6.2.3. Growth through expanding varieties

With symmetric quantities, $X_{it} = X_t$ for all i. Define total intermediate use as M_t:

$$\begin{aligned} M_t &= \int_0^{N_t} X_{it}\, \mathrm{d}i = N_t X_t, \\ Y_t &= A L^{1-\alpha} N_t X_t^{\alpha} \\ &= A L^{1-\alpha} M_t^{\alpha} N_t^{1-\alpha}. \end{aligned} \tag{6.4}$$

- At fixed L and M_t, **more varieties raise output** because $1-\alpha > 0$. Each variety has diminishing returns; spreading a given total input across more varieties raises production.
- At fixed L and quantity per variety X_t, output is **linear in** N_t. Variety expansion can sustain growth without increasing A.
- Old varieties remain in use: this model has **no creative destruction**.

Lecture addition: Output exhaustion under competitive factor pricing

In the symmetric representation, write $Y = F(L, M, N)$. Then

$$\begin{aligned} L F_L + M F_M &= Y, \\ N F_N &= (1-\alpha) Y > 0. \end{aligned}$$

Paying labor and physical inputs their marginal products already exhausts output. Paying knowledge its positive marginal product as well would require payments exceeding output.

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6.2.4. Problem of the final-goods firm

The representative final-goods firm takes wage w_t and intermediate prices P_{it} as given. Write its labor demand as ℓ_t; labor-market clearing requires $\ell_t = L$.

Each period it chooses nonnegative inputs to solve

$$\max_{\ell_t, \{X_{it}\}} \left[A \ell_t^{1-\alpha} \int_0^{N_t} X_{it}^{\alpha}\, \mathrm{d}i - w_t \ell_t - \int_0^{N_t} P_{it} X_{it}\, \mathrm{d}i \right]. \tag{6.5}$$

The interior FOCs, evaluated at $\ell_t = L$, are

$$\begin{aligned} w_t &= (1-\alpha)\frac{Y_t}{L}, \\ P_{it} &= A\alpha L^{1-\alpha} X_{it}^{\alpha-1} = A\alpha\left(\frac{L}{X_{it}}\right)^{1-\alpha}. \end{aligned} \tag{6.6}$$

Each input is paid its **marginal product**. Solving the second FOC gives the demand faced by intermediate producer i:

$$X_{it} = L\left(\frac{A\alpha}{P_{it}}\right)^{\frac{1}{1-\alpha}}. \tag{6.7}$$

Demand has constant price elasticity $-\frac{1}{1-\alpha}$. For given N_t, constant returns and price taking imply zero final-goods profits.

6.2.5. Problem of an intermediate-goods firm

Each firm holds a permanent patent on one variety and has marginal cost one. The initial R&D cost is sunk. Pricing is **static**: today's price does not affect future demand.

Let $\varepsilon = \frac{1}{1-\alpha} > 1$ be the absolute demand elasticity. The firm chooses its price to maximize operating profit:

$$\max_{P>0}(P-1)X(P), \quad X(P) = L\left(\frac{A\alpha}{P}\right)^{\varepsilon}. \tag{6.8}$$

Since $X'(P) = -\varepsilon\frac{X(P)}{P}$, the FOC gives

$$\begin{aligned} 1 - \varepsilon\frac{P-1}{P} &= 0, \\ P = \frac{\varepsilon}{\varepsilon-1} &= \frac{1}{\alpha} > 1. \end{aligned} \tag{6.9}$$

The price is a **constant markup** over marginal cost. Quantity and per-period operating profit are

$$\begin{aligned} X &= L\left(A\alpha^2\right)^{\frac{1}{1-\alpha}}, \\ \pi &= (P-1)X = \frac{1-\alpha}{\alpha} L\left(A\alpha^2\right)^{\frac{1}{1-\alpha}}. \end{aligned} \tag{6.10}$$

With A, L and α constant, P, X and π are constant across varieties and over time. These profits exclude the initial R&D outlay.

Hold N_t fixed. Efficient production uses each intermediate until its marginal product falls to its production cost of one:

$$\begin{aligned} X^* &= L(A\alpha)^{\frac{1}{1-\alpha}}, \\ \frac{X}{X^*} &= \alpha^{\frac{1}{1-\alpha}} < 1. \end{aligned} \tag{6.11}$$

The second line of Equation (6.11) shows that the monopoly markup leaves each variety's quantity below the efficient level X^*, holding N_t fixed.

6.2.6. Value of a patent and free entry

A patent created in t earns π each period from $t+1$ onward. With a constant net interest rate $r > 0$, its value at t is

$$\begin{aligned} V &= \sum_{s=1}^{\infty} \frac{\pi}{(1+r)^s} \\ &= \pi \frac{\frac{1}{1+r}}{1 - \frac{1}{1+r}} = \frac{\pi}{r}. \end{aligned} \tag{6.12}$$

Inventing a patent costs η, so net entry profit is $V - \eta$.

- If $V > \eta$, the linear R&D technology implies unbounded desired entry at those prices. This cannot be an equilibrium.
- If $V < \eta$, no firm invests in R&D.
- With positive R&D, **free entry** requires $V = \eta$. Net entry profit is zero, while the patent continues to earn positive operating profits π.

The equilibrium conditions allow zero R&D even when $V = \eta$:

$$V \le \eta, \quad R_t \ge 0, \quad (\eta - V) R_t = 0. \tag{6.13}$$

With **positive innovation**, $V = \eta = \frac{\pi}{r}$, hence

$$r = \frac{\pi}{\eta} = \frac{1-\alpha}{\alpha} \frac{L}{\eta} \left(A\alpha^2\right)^{\frac{1}{1-\alpha}}. \tag{6.14}$$

6.2.7. The household

The representative household supplies L inelastically and owns claims on patents. It takes wages and asset returns as given. Given a_0, its problem is

$$\max_{\{c_t, a_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad \beta = \frac{1}{1+\rho}, \tag{6.15}$$

with $\rho > 0$ and CRRA utility for $c > 0$:

$$u(c) = \begin{cases} \frac{c^{1-\sigma}}{1-\sigma} & \text{if } \sigma \neq 1 \\ \log c & \text{if } \sigma = 1 \end{cases}, \quad \sigma > 0. \tag{6.16}$$

Here ρ is the rate of time preference and $\frac{1}{\sigma}$ is the intertemporal elasticity of substitution. With one representative household, $c_t = C_t$.

The household faces a no-Ponzi condition and the budget constraint

$$c_t + a_{t+1} = (1 + r_t) a_t + w_t L. \tag{6.17}$$

Assets a_t were chosen in $t-1$ and earn the return r_t in period t; saving a_{t+1} earns r_{t+1}.

A patent bought for V_t at the end of t pays π and can be resold for V_{t+1} next period. Its **total return** includes both profits and capital gains:

$$1 + r_{t+1} = \frac{\pi + V_{t+1}}{V_t}. \tag{6.18}$$

When the patent value is constant, $1 + r = \frac{\pi+V}{V}$. Patent profits are already included in the asset return in the budget.

At an interior optimum, saving equates the marginal utility cost today to the discounted marginal utility of its payoff:

$$\begin{aligned} u'(c_t) &= \beta(1 + r_{t+1})u'(c_{t+1}), \\ \left(\frac{c_{t+1}}{c_t}\right)^{\sigma} &= \beta(1 + r_{t+1}). \end{aligned} \tag{6.19}$$

6.2.8. Equilibrium growth

With **positive innovation**, free entry fixes $V = \eta$ and $r = \frac{\pi}{\eta}$. Substituting this return into the household Euler equation gives

$$\begin{aligned} 1 + g \coloneqq \frac{C_{t+1}}{C_t} &= [\beta(1 + r)]^{\frac{1}{\sigma}} \\ &= \left(\frac{1 + \frac{\pi}{\eta}}{1 + \rho}\right)^{\frac{1}{\sigma}}. \end{aligned} \tag{6.20}$$

Positive growth requires $r > \rho$. If $\frac{\pi}{\eta} \leq \rho$, the stationary corner has $R_t = 0$, $r = \rho$ and $V = \frac{\pi}{\rho} \leq \eta$.

On the positive-innovation path, **asset-market clearing** and **goods-market clearing** require

$$\begin{aligned} a_{t+1} &= V_t N_{t+1} = \eta N_{t+1}, \\ Y_t &= C_t + N_t X + \eta(N_{t+1} - N_t). \end{aligned} \tag{6.21}$$

End-of-period assets include new patents, whose first profits arrive in $t + 1$.

On the BGP, $N_{t+1} = (1 + g)N_t$. With A and X constant, output, consumption and R&D grow at rate g:

$$\begin{aligned} \frac{Y_t}{N_t} &= AL^{1-\alpha}X^{\alpha}, \\ \frac{R_t}{N_t} &= \eta g, \\ \frac{C_t}{N_t} &= AL^{1-\alpha}X^{\alpha} - X - \eta g > 0. \end{aligned} \tag{6.22}$$

Lecture addition: Why indifference still determines aggregate R&D

Free entry leaves each potential entrant indifferent. A higher proposed BGP entry rate raises consumption growth. The Euler equation would then require a higher r, but this would lower $V = \frac{\pi}{r}$ below the invention cost η. Household saving and goods-market clearing select the aggregate entry mass.

An extension with research labor and heterogeneous tastes for innovating could yield a smooth aggregate supply of innovation.

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Lecture addition: Why there is no transition in this model

The only aggregate production state is N_t. Its initial level scales allocations while leaving π, r and g unchanged. In the growing equilibrium, consumption adjusts immediately to its BGP level relative to N_0.

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Since assets grow at rate g, household transversality requires

$$\beta(1+g)^{1-\sigma} = \frac{1+g}{1+r} < 1 \quad \Leftrightarrow \quad g < r. \tag{6.23}$$

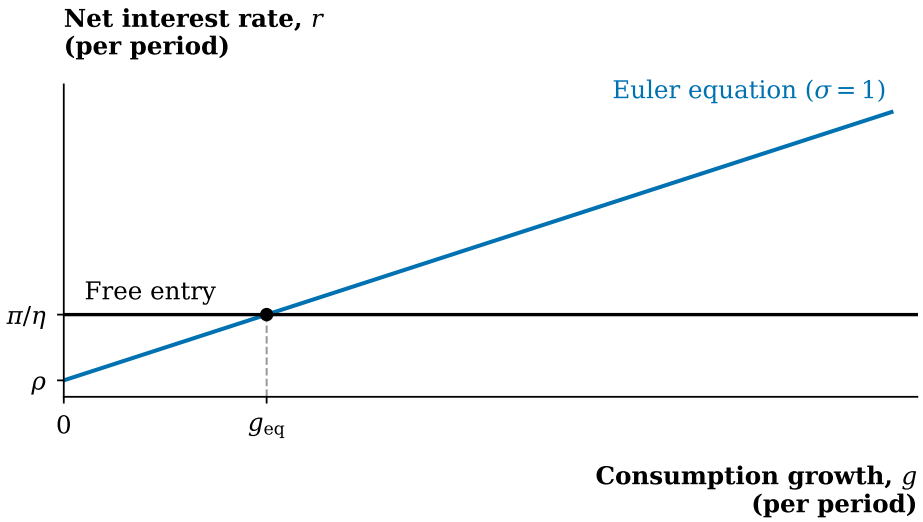


Figure 14: Equilibrium growth.

Note: Log-utility schematic ($\sigma = 1$), redrawn from Lecture 6, slide 11. **Free entry** fixes the return on R&D at $r = \frac{\pi}{\eta}$. The **Euler equation**, $r = (1+\rho)(1+g)^{\sigma} - 1$, maps this return into consumption growth g_{eq} at the intersection.

6.2.9. Determinants of growth

From Equation (6.20), holding other parameters fixed:

- **Higher productivity** A raises patent profits and growth.
- **A larger economy** L raises profits and growth through a larger market (*scale effect*).
- **A higher innovation cost** η lowers the return $r = \frac{\pi}{\eta}$ and growth.
- **Greater impatience** ρ lowers saving and growth at a given return.

Lecture addition: Scale and the market for one variety

Increasing L expands the market for each variety. Increasing N_t adds varieties at unchanged demand per variety, so N_t itself does not raise the growth rate. With market integration, innovators can serve customers beyond national borders; national population alone need not measure their market size.

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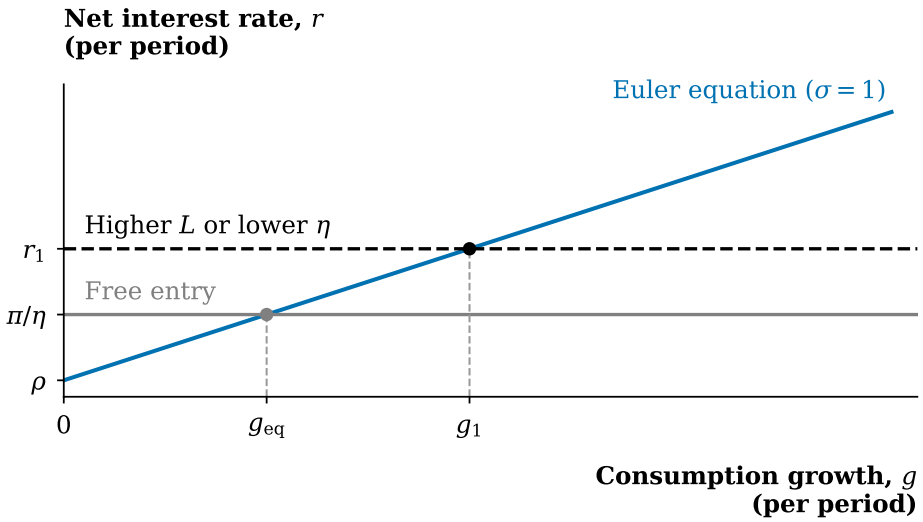


Figure 15: A larger market or cheaper innovation.

Note: Log-utility schematic ($\sigma = 1$), redrawn from Lecture 6, slide 11. Higher L or lower η raises $\frac{\pi}{\eta}$, shifting the free-entry line upward to r_1. Preferences are unchanged, so the Euler locus stays fixed and growth rises from g_{eq} to g_1.

6.2.10. The planner's problem

Given $N_0 > 0$, the planner has the same preferences as the household and solves

$$\max_{\{C_t, X_t, N_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(C_t) \tag{6.24}$$

subject to

$$C_t + N_t X_t + \eta(N_{t+1} - N_t) = A L^{1-\alpha} N_t X_t^{\alpha}, \tag{6.25}$$

with $C_t > 0$, $X_t \geq 0$ and $N_{t+1} \geq N_t$.

Static choice: for given N_t, use each variety until its marginal product equals its marginal cost:

$$\begin{aligned} A\alpha L^{1-\alpha}(X^*)^{\alpha-1} &= 1, \\ X^* &= L(A\alpha)^{\frac{1}{1-\alpha}} > X. \end{aligned} \tag{6.26}$$

The **social return to innovation** is net output from a new variety divided by its cost. Using the static FOC,

$$\begin{aligned} r^* &= \frac{AL^{1-\alpha}(X^*)^{\alpha} - X^*}{\eta} \\ &= \frac{1-\alpha}{\alpha\eta}X^* = \frac{1-\alpha}{\alpha\eta}L(A\alpha)^{\frac{1}{1-\alpha}}. \end{aligned} \tag{6.27}$$

Blueprints persist, so the gross return is $1 + r^*$. The resource constraint becomes

$$C_t + \eta N_{t+1} = (1 + r^*)\eta N_t. \tag{6.28}$$

Dynamic choice: investing η today adds a variety next period. At an interior optimum,

$$\begin{aligned} u'(C_t) &= \beta(1 + r^*)u'(C_{t+1}), \\ 1 + g^* &:= \frac{C_{t+1}}{C_t} = \left(\frac{1 + r^*}{1 + \rho}\right)^{\frac{1}{\sigma}}. \end{aligned} \tag{6.29}$$

An interior growing path requires $r^* > \rho$ and $g^* < r^*$.

The markup reduces intermediate use and private innovation returns. When both paths have positive innovation, **equilibrium growth is too low**:

$$\frac{r}{r^*} = \frac{X}{X^*} = \alpha^{\frac{1}{1-\alpha}} < 1 \quad \Rightarrow \quad g < g^*. \tag{6.30}$$

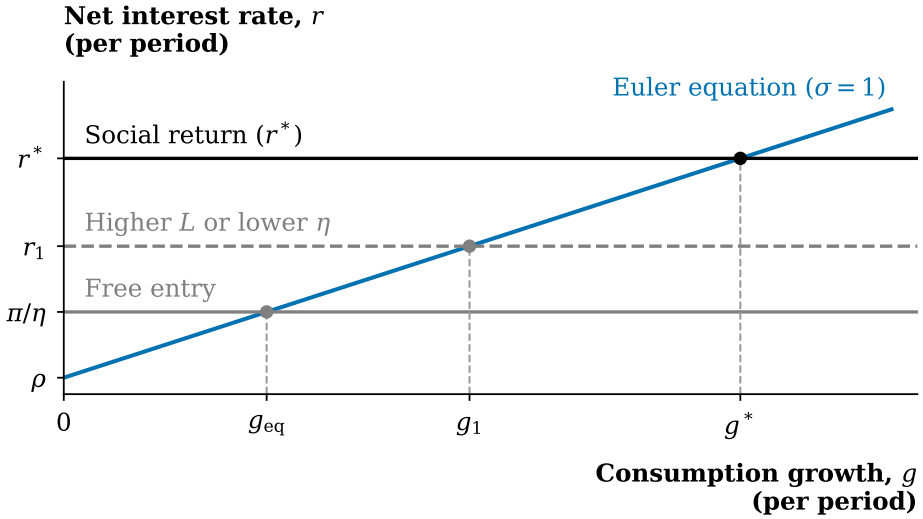


Figure 16: Equilibrium and planner growth.

Note: Log-utility schematic ($\sigma = 1$), redrawn from Lecture 6, slide 11. The planner comparison holds the baseline parameters fixed: $r^* > \frac{\pi}{\eta}$ implies $g^* > g_{\text{eq}}$. The middle line repeats the higher-L or lower-η example above.

6.2.11. Policy implications

In this model:

- **No patent protection:** free imitation forces $P = 1$, so $\pi = V = 0$. Firms cannot recover the innovation cost $\eta > 0$. Private R&D stops, giving $g = 0$.

- **Finite patent protection:** a patent lasting T periods has value

$$V(T) = \sum_{s=1}^{T} \frac{\pi}{(1+r)^s} = \frac{\pi}{r}\left[1 - (1+r)^{-T}\right] < \frac{\pi}{r}. \tag{6.31}$$

  At given $r > 0$ and π, shorter protection lowers the reward to innovation. Free entry then supports a lower return and growth rate than with permanent patents, or no R&D if protection is too short.
- **Subsidize intermediate purchases:** with subsidy rate s_X, the buyer pays $(1 - s_X)P$. Since producers charge $P = \frac{1}{\alpha}$, setting $s_X = 1 - \alpha$ makes the buyer's price equal marginal cost, one. This restores X^* while preserving producers' innovation incentives. Profits become $\pi^* = \frac{1-\alpha}{\alpha} X^*$, so free entry gives $r = \frac{\pi^*}{\eta} = r^*$ and implements planner growth.
- **Subsidize R&D:** a subsidy rate s_R lowers the private invention cost to $(1 - s_R)\eta$. With positive innovation, $r = \frac{\pi}{(1-s_R)\eta}$, raising growth. This can reduce underinvestment, but the monopoly markup and low intermediate use remain.

The subsidy comparisons assume **lump-sum financing**.

6.3. Quality ladders: main idea

(Aghion & Howitt, 1992): *Growth comes from successive improvements in the quality of existing products.*

- **Quality improvements:** the number of sectors is fixed. A successful innovation in sector j raises quality from A_{jt} to γA_{jt}, where $\gamma > 1$.
- **Creative destruction:** the innovator becomes the new monopolist, displacing the incumbent. The previous leader loses its future monopoly rents.
- **Business stealing:** part of the entrant's reward is taking over the incumbent's rents. This transfer is a private gain without a corresponding net gain to society, which can encourage excessive R&D.

Quality gains persist after the innovator is replaced, so its profits capture only part of their social value. This can discourage R&D. Together with business stealing, this means **equilibrium innovation can be too high or too low** relative to the planner's choice.

6.3.1. (Aghion & Howitt, 1992) in discrete time: environment

Time is discrete. There is a fixed continuum of intermediate sectors $j \in [0, 1]$ and labor supply $L > 0$.

Final-good production is competitive, with the final good as numeraire:

$$Y_t = L^{1-\alpha} \int_0^1 A_{jt}^{1-\alpha} x_{jt}^{\alpha} \, \mathrm{d}j, \quad 0 < \alpha < 1. \tag{6.32}$$

Here $A_{jt} > 0$ is the leading quality in sector j, and $x_{jt} \geq 0$ is the quantity of its intermediate input.

Intermediate production: the current leader produces one unit of x_{jt} from one unit of the final good, so marginal cost is one. Quality steps are assumed large enough that older versions do not constrain the leader's monopoly price.

Innovation: entrants spend $R_{jt} \geq 0$ units of the final good on research. The probability of a quality improvement is

$$\mu_{jt} = \frac{\lambda R_{jt}}{\gamma A_{jt}} \in [0,1], \quad \lambda > 0, \quad \gamma > 1. \tag{6.33}$$

The parameter λ measures research productivity; γ is the quality step. Feasible spending satisfies $R_{jt} \leq \gamma \frac{A_{jt}}{\lambda}$. Quality evolves according to

$$A_{j,t+1} = \begin{cases} \gamma A_{jt} & \text{with probability } \mu_{jt} \\ A_{jt} & \text{with probability } 1 - \mu_{jt} \end{cases}. \tag{6.34}$$

Current leaders produce in t; research outcomes arrive at the end of t. A successful entrant replaces the incumbent and starts producing in $t+1$. Without innovation, the incumbent remains the leader.

Resources and ownership: the representative household supplies L inelastically, owns firms and has the same CRRA preferences as above. Final output covers consumption, intermediate production and research:

$$Y_t = C_t + \underbrace{\int_0^1 x_{jt}\,\mathrm{d}j}_{\text{intermediate production}} + \underbrace{\int_0^1 R_{jt}\,\mathrm{d}j}_{\text{R\&D}}. \tag{6.35}$$

6.3.2. Monopoly profits

Intermediate demand: the competitive final-good firm pays each input its marginal product. If P_{jt} is the price of intermediate j,

$$\begin{aligned} P_{jt} &= \alpha L^{1-\alpha} A_{jt}^{1-\alpha} x_{jt}^{\alpha-1}, \\ x_{jt} &= A_{jt} L \left(\frac{\alpha}{P_{jt}} \right)^{\frac{1}{1-\alpha}}. \end{aligned} \tag{6.36}$$

Quality scales demand; its absolute price elasticity remains $\varepsilon = \frac{1}{1-\alpha} > 1$.

Monopoly pricing: with marginal cost one, the leader maximizes $(P-1)x_{jt}(P)$. The constant-elasticity markup gives

$$P_{jt} = \frac{\varepsilon}{\varepsilon - 1} = \frac{1}{\alpha} > 1. \tag{6.37}$$

This uses the assumption that older versions do not constrain the price. Substituting into demand gives quantity and **operating profit**:

$$
\begin{aligned}
x_{jt} &= \alpha^{\frac{2}{1-\alpha}} A_{jt} L, \\
\pi_{jt} &= \big(P_{jt} - 1\big) x_{jt} = \frac{1-\alpha}{\alpha} x_{jt} = \overline{\pi} A_{jt} L, \\
\overline{\pi} &\coloneqq (1-\alpha)\alpha^{\frac{1+\alpha}{1-\alpha}}.
\end{aligned}
\tag{6.38}
$$

Profits exclude the initial R&D cost. A quality upgrade by factor γ raises per-period profit by the same factor while the innovator remains the leader.

Aggregate output: define average quality over the unit mass of sectors as $A_t \coloneqq \int_0^1 A_{jt}\,\mathrm{d}j$. Substituting equilibrium quantities,

$$
\begin{aligned}
Y_t &= L^{1-\alpha} \int_0^1 A_{jt}^{1-\alpha} \left[\alpha^{\frac{2}{1-\alpha}} A_{jt} L\right]^{\alpha} \mathrm{d}j \\
&= \alpha^{\frac{2\alpha}{1-\alpha}} A_t L.
\end{aligned}
\tag{6.39}
$$

With L fixed, aggregate output grows at the same rate as average quality.

6.3.3. Creative destruction and the value of innovation

Innovation value: take $r > 0$ and the common replacement probability $\mu \in [0,1]$ as constant. Let $V(A')$ be the end-of-period value of a leader with quality A', before its first profit next period. It first receives $\overline{\pi} A' L$, then retains its continuation value with probability $1-\mu$. Replacement leaves zero continuation value:

$$
\begin{aligned}
V(A') &= \frac{\overbrace{\overline{\pi} A' L}^{\text{next-period profit}} + \overbrace{(1-\mu) V(A')}^{\text{surviving leader}}}{1+r}, \\
(r+\mu) V(A') &= \overline{\pi} A' L, \\
V(A') &= \frac{\overline{\pi} A' L}{r+\mu}.
\end{aligned}
\tag{6.40}
$$

The **effective discount rate** is $r + \mu$. Faster future innovation shortens the profit stream and lowers the patent's value, holding A' and r fixed.

> **Lecture addition: A patent's legal and economic life**
>
> Legal protection can outlast a product's economic life. An incumbent may retain the right to sell its old technology after customers have switched to a better version. The displacement probability μ ends its profit stream through obsolescence.
>
> Recording: 2026-10-19, 00:58:39 - 00:59:22

Free entry: spending R_{jt} in t gives probability $\mu_{jt} = \lambda \frac{R_{jt}}{\gamma A_{jt}}$ of obtaining a new patent with end-of-t value $V\big(\gamma A_{jt}\big)$. Expected net entry profit is

$$
\mu_{jt} V\big(\gamma A_{jt}\big) - R_{jt} = R_{jt}\left[\frac{\lambda V\big(\gamma A_{jt}\big)}{\gamma A_{jt}} - 1\right].
\tag{6.41}
$$

For **interior positive R&D**, free entry sets the expected return per unit spent equal to one:

$$\begin{aligned} \frac{\lambda V(\gamma A_{jt})}{\gamma A_{jt}} = \frac{\lambda \overline{\pi} L}{r+\mu} = 1, \\ r + \mu = \lambda \overline{\pi} L. \end{aligned} \tag{6.42}$$

Quality cancels because both patent value and the cost of a given success probability scale with γA_{jt}. The **free-entry locus** is $r = \lambda\overline{\pi}L - \mu$: a higher innovation rate lowers the interest rate consistent with entry breaking even.

6.3.4. Equilibrium growth

Aggregation: with a common innovation probability μ and diversified sectoral risk, aggregate quality evolves as

$$\begin{aligned} A_{t+1} &= \int_0^1 \left[(1-\mu)A_{jt} + \mu\gamma A_{jt}\right] \mathrm{d}j \\ &= [1 + \mu(\gamma - 1)]A_t, \\ g &:= \frac{A_{t+1}}{A_t} - 1 = \mu(\gamma - 1). \end{aligned} \tag{6.43}$$

On a BGP, research spending and consumption are proportional to A_t:

$$\begin{aligned} R_t &:= \int_0^1 R_{jt}\,\mathrm{d}j = \frac{\gamma\mu}{\lambda} A_t, \\ \frac{C_t}{A_t} &= L\left[\alpha^{\frac{2\alpha}{1-\alpha}} - \alpha^{\frac{2}{1-\alpha}}\right] - \frac{\gamma\mu}{\lambda} > 0. \end{aligned} \tag{6.44}$$

Thus quality, output and consumption all grow at rate g.

Equilibrium conditions: the household Euler equation and free entry give

$$\begin{aligned} [1 + \mu(\gamma - 1)]^\sigma &= \frac{1+r}{1+\rho}, \\ r &= \lambda\overline{\pi}L - \mu. \end{aligned} \tag{6.45}$$

Combining them determines μ:

$$(1+\rho)[1 + \mu(\gamma - 1)]^\sigma = 1 + \lambda\overline{\pi}L - \mu. \tag{6.46}$$

The left side increases in μ; the right side decreases. Any interior intersection is unique, with $\rho < \lambda\overline{\pi}L < (1+\rho)\gamma^\sigma$ ensuring $0 < \mu < 1$. The BGP must also satisfy household transversality, $g < r$.

Log utility ($\sigma = 1$) gives

$$
\begin{aligned}
\mu[1+(1+\rho)(\gamma-1)] &= \lambda\bar{\pi}L-\rho,\\
\mu &= \frac{\lambda\bar{\pi}L-\rho}{1+(1+\rho)(\gamma-1)},\\
g &= (\gamma-1)\frac{\lambda\bar{\pi}L-\rho}{1+(1+\rho)(\gamma-1)}.
\end{aligned}
\tag{6.47}
$$

If $\lambda\bar{\pi}L \le \rho$, the **no-innovation corner** has $\mu = g = 0$ and $r = \rho$; the positive-entry equality no longer applies.

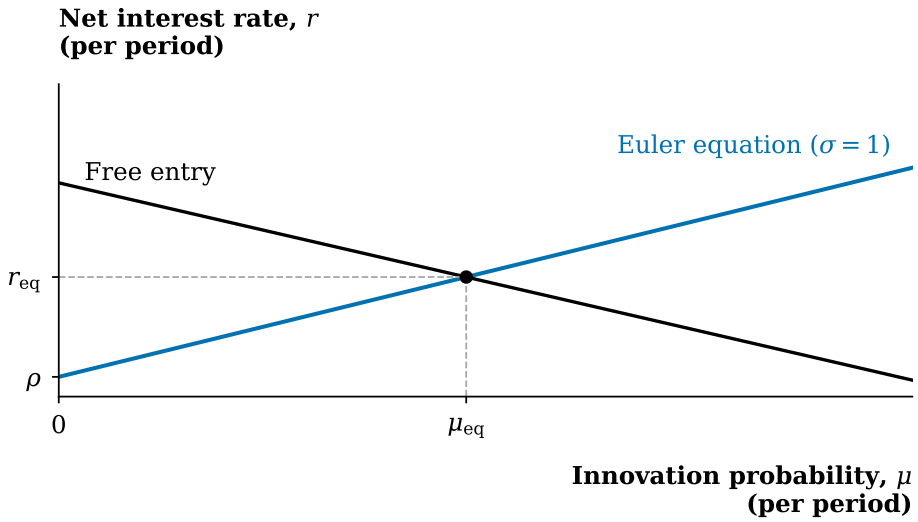


Figure 17: Equilibrium innovation.

Note: Log-utility schematic ($\sigma = 1$), redrawn from Lecture 6, slide 21. Free entry gives the downward-sloping locus $r = \lambda\bar{\pi}L - \mu$. The Euler locus $r = (1+\rho)[1+\mu(\gamma-1)]^{\sigma} - 1$ slopes upward. Their intersection determines μ_{eq} and r_{eq}.

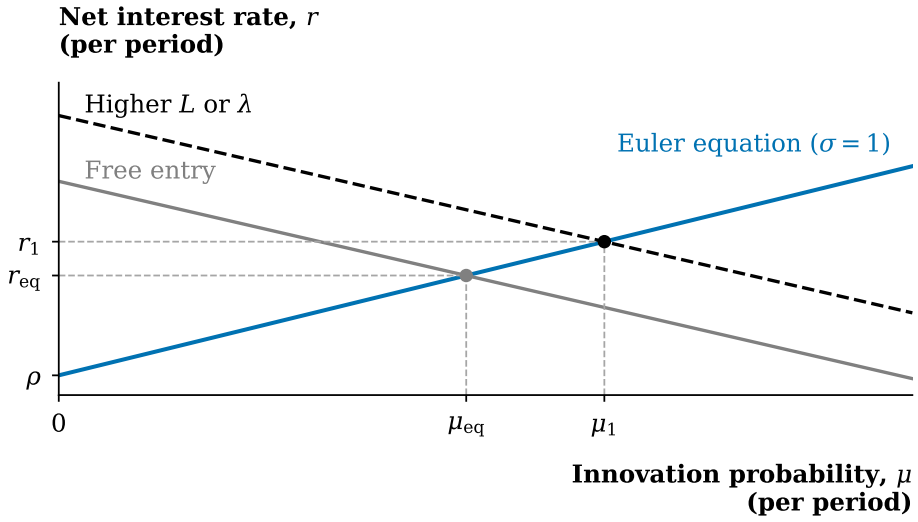


Figure 18: Market size and research productivity.

Note: Log-utility schematic ($\sigma = 1$), redrawn from Lecture 6, slide 21. Higher L or λ shifts the free-entry locus upward; preferences and γ are unchanged. The new intersection has higher μ and r, raising growth through $g = \mu(\gamma-1)$.

6.3.5. Determinants of growth

Using $\mu = \frac{g}{\gamma-1}$, equilibrium growth satisfies

$$(1+\rho)(1+g)^{\sigma} + \frac{g}{\gamma - 1} = 1 + \lambda \bar{\pi} L. \tag{6.48}$$

The left side increases in g. Within an **interior positive-growth equilibrium**, holding other parameters fixed:

- **A larger economy** L raises monopoly profits and shifts free entry upward, increasing μ and g. This is the *scale effect*.
- **Higher research productivity** λ raises the success probability per unit of research spending, increasing μ and g.
- **Larger quality steps** γ raise the gain from each successful innovation. Growth g rises even though innovation becomes less frequent: the Euler equation implies higher r, so free entry requires lower μ.
- **More patient households** (lower ρ) shift the Euler locus downward, increasing μ and g.

Sectoral quality rises in jumps. Successful entrants replace incumbents and take over production. Smooth aggregate growth can coexist with **firm entry, exit and reallocation**.

Lecture addition: Common research rates, different outcomes

Sectors face the same μ, but their realized sequences of successes differ. Some firms are displaced quickly; others survive for many periods. Sectoral quality and firm ages can therefore differ under a common research technology.

Recording: 2026-10-19, 01:06:56 - 01:07:57

6.3.6. Is growth too fast or too slow?

Private rewards to innovation differ from social returns in three ways:

- **Appropriability:** the innovator captures monopoly profits, but not the full surplus generated by higher quality. This pushes toward **too little innovation**.
- **Intertemporal spillover:** future innovators build on today's quality. The current innovator receives no payment for these gains after being replaced, pushing toward **too little innovation**.
- **Business stealing:** the entrant gains rents that the incumbent loses. This transfer adds no social surplus. Ignoring the incumbent's loss pushes toward **too much innovation**.

The net effect is ambiguous: equilibrium innovation and growth can be above or below the planner's choice. When quality improvements are small relative to the rents captured, business stealing can dominate and growth can be too fast.

Lecture addition: Innovation policy under full information

These incentive wedges exist with commonly known research costs and technology. Which wedge dominates must be assessed sector by sector. Small persistent changes in growth compound into large differences in future consumption.

Private information about research costs or opportunities would add another policy constraint.

Recording: 2026-10-19, 01:10:42 - 01:14:04

6.4. Where has the literature gone since?

> **Lecture addition: Growth and industrial organization**
>
> The lecture emphasized detailed empirical extensions of these baselines, using firm-, industry- and sector-level data to study innovation incentives. This connects growth theory with **industrial organization**.
>
> Recording: 2026-10-19, 01:14:04 - 01:16:01

Supplementary reading.

- **Scale effects and semi-endogenous growth.** (Jones, 1995): *The scale-effect prediction conflicts with time-series evidence. A semi-endogenous extension makes long-run growth depend on population growth, even though technological progress comes from R&D.*
- **Firm dynamics.** (Klette & Kortum, 2004): *Heterogeneous firms innovate, enter and exit. Their model connects firm-level evidence on R&D and patenting to aggregate innovation and growth.*
- **Competition and innovation.** (Aghion et al., 2005): *Competition can encourage neck-and-neck firms to innovate and discourage laggards. Their model and panel evidence yield an inverted-U relationship between competition and innovation.*
- **The direction of innovation.** (Acemoglu et al., 2012): *Research can improve clean or dirty technologies. Carbon prices and research subsidies redirect innovation; their model's optimal policy uses both instruments.*
- **Research productivity.** (Bloom et al., 2020): *They document falling measured research productivity across several industries and firms. Maintaining a given rate of technological improvement requires increasing research inputs in these cases.*
- **Innovation policy with heterogeneous firms.** (Akcigit et al., 2022): *Firms differ in research productivity, which the government cannot observe directly. Information constraints change the taxes and R&D subsidies needed to address knowledge spillovers.*

6.5. Scale effects and semi-endogenous growth

Empirical motivation. (Jones, 1995): *Research employment rose in industrialized economies without the growth acceleration predicted by the early models' strong scale effects.*

6.5.1. Idea production

Use a discrete-time version of the idea-production block in (Jones, 2005):

$$A_{t+1} - A_t = \chi L_{A,t}^{\eta} A_t^{\varphi}, \quad \chi > 0, \quad 0 < \eta \leq 1, \quad 0 \leq \varphi < 1. \tag{6.49}$$

- $A_t > 0$: stock of ideas; $L_{A,t}$: researchers.
- χ: research efficiency; η: research-labor elasticity; φ: elasticity with respect to existing knowledge.
- Take a constant equilibrium research share $0 < s_A < 1$ and population growth $n > 0$: $L_{A,t} = s_A L_t$, $L_{t+1} = (1+n)L_t$.

The proportional growth rate of ideas is

$$g_{A,t} \coloneqq \frac{A_{t+1} - A_t}{A_t} = \frac{\chi(s_A L_t)^\eta}{A_t^{1-\varphi}}. \tag{6.50}$$

With $\varphi = 1$, growth rises with the number of researchers. With $\varphi < 1$, a larger idea stock reduces proportional growth at a given research input.

6.5.2. Balanced growth

With s_A fixed,

$$\frac{g_{A,t+1}}{g_{A,t}} = \frac{(1+n)^\eta}{\left(1 + g_{A,t}\right)^{1-\varphi}}. \tag{6.51}$$

On a BGP, $g_{A,t+1} = g_{A,t} = g_A > 0$, so

$$\begin{aligned} \left(1 + g_A\right)^{1-\varphi} &= (1+n)^\eta, \\ g_A &= (1+n)^{\frac{\eta}{1-\varphi}} - 1 \approx \frac{\eta n}{1-\varphi}. \end{aligned} \tag{6.52}$$

The approximation applies to small growth rates. This is **semi-endogenous growth**: R&D produces new ideas, while exogenous population growth and production elasticities pin down the BGP rate. With $n = 0$, ideas can keep accumulating, but their proportional growth rate tends to zero.

6.5.3. Levels and policy

For $n > 0$, the BGP satisfies

$$\frac{A_t}{L_t^{\frac{\eta}{1-\varphi}}} = \left(\frac{\chi s_A^\eta}{g_A}\right)^{\frac{1}{1-\varphi}}. \tag{6.53}$$

- **A level scale effect remains:** a larger population raises the BGP idea stock, holding its growth rate n fixed.
- **A permanent rise in s_A** raises growth during the transition and shifts the technology path upward. The BGP growth rate is unchanged. Policies affecting research allocation can change income levels without changing long-run growth, holding n, η and φ fixed.
- **Relevant scale:** L_t is the population sharing the idea stock, potentially global when ideas cross borders.

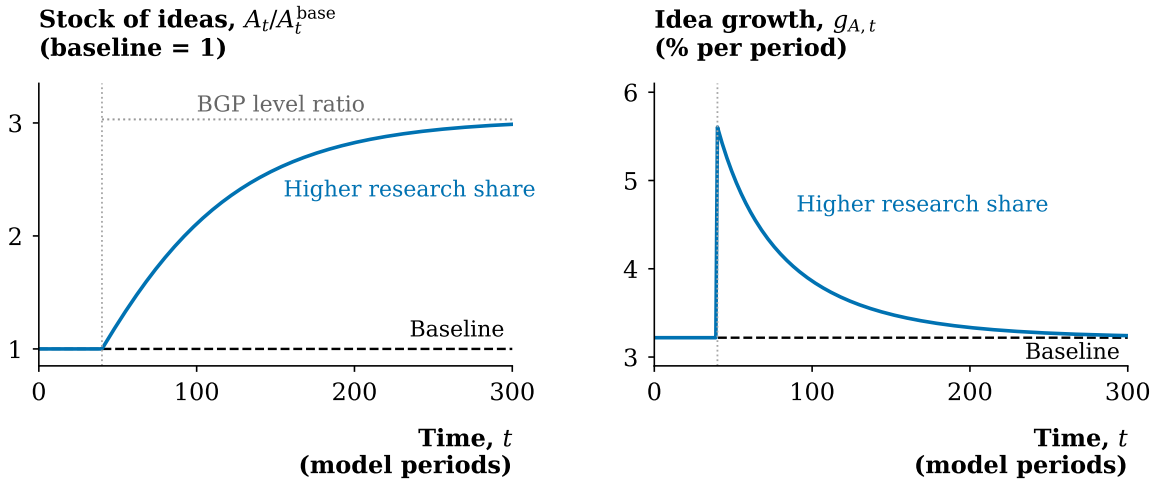


Figure 19: Research intensity: levels and growth.

Note: Uncalibrated example with $n = 0.02$, $\eta = 0.8$, $\varphi = 0.5$ and $A_0 = L_0 = 1$. Research share rises from 0.05 to 0.10 in period 40; χ puts the initial economy on its BGP. The idea-stock ratio tends to $2^{\frac{\eta}{1-\varphi}} \approx 3.03$, while growth returns to $(1.02)^{1.6} - 1 \approx 3.22\%$ per period.

6.6. Are ideas harder to find?

(Bloom et al., 2020): *Maintaining a given rate of technological improvement requires rising research inputs in their case studies.*

6.6.1. Measuring research productivity

Let S_t denote effective research input. Define

$$\mathcal{P}_t := \frac{g_{A,t}}{S_t}, \quad g_{A,t} := \frac{A_{t+1} - A_t}{A_t}. \tag{6.54}$$

Output is a proportional improvement, such as TFP growth or chip-density growth. Effective researchers are research spending divided by a wage proxy, earnings of college-educated men, to adjust for rising researcher costs.

6.6.2. Evidence

- **US aggregate, since the 1930s:** *Effective research input rose about 23-fold; research productivity fell about 41-fold. Output is measured by decadal TFP growth.*
- **Moore's law, 1971–2014:** *Chip density roughly doubled every two years. Research input rose about 18-fold; research productivity fell about 6.8% per year.*[2]
- **Other cases:** *Crop yields, medical advances and firm outcomes generally show declining research productivity, with variation across cases and measures.*

[2]The lecture used roughly eighteen months. This paragraph follows the paper's approximately two-year transistor-density benchmark (recording, 2026-10-19, 01:19:44 - 01:20:10).

6.6.3. Connection to semi-endogenous growth

In the preceding model, taking $S_t = L_{A,t}$ gives

$$\mathcal{R}_t = \chi S_t^{\eta-1} A_t^{\varphi-1}. \tag{6.55}$$

At fixed S_t, $\varphi < 1$ makes research productivity fall as knowledge accumulates. Maintaining constant positive g_A therefore requires growing research effort. Falling measured productivity is consistent with this mechanism; the ratio alone does not identify φ.

Lecture addition: AI and the research technology

The lecturer left open how much AI changes research technology. In the semi-endogenous block above, a permanent rise in χ alone raises BGP technology levels while leaving the long-run growth rate g_A unchanged.

Recording: 2026-10-19, 01:19:11 - 01:19:29; 01:21:13 - 01:21:25

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